

Hydrogeological Atlas of Rajasthan

Other Nallahs Basin



2013

Message



Shri Ashok Gehlot,
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of '*Hydrogeological Atlas of Rajasthan*'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.



(Ashok Gehlot)

Message



Dr. Jitendra Singh,
Hon'ble Minister for Energy,
Non-Conventional Energy Resources,
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.



15.4.13.

(Jitendra Singh)

Message



Dr. Purushottam Agarwal

Principal Secretary to Government,

Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of '*Hydrogeological Atlas of Rajasthan*' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

(Purushottam Agarwal)

Foreword



Arno Schaefer

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

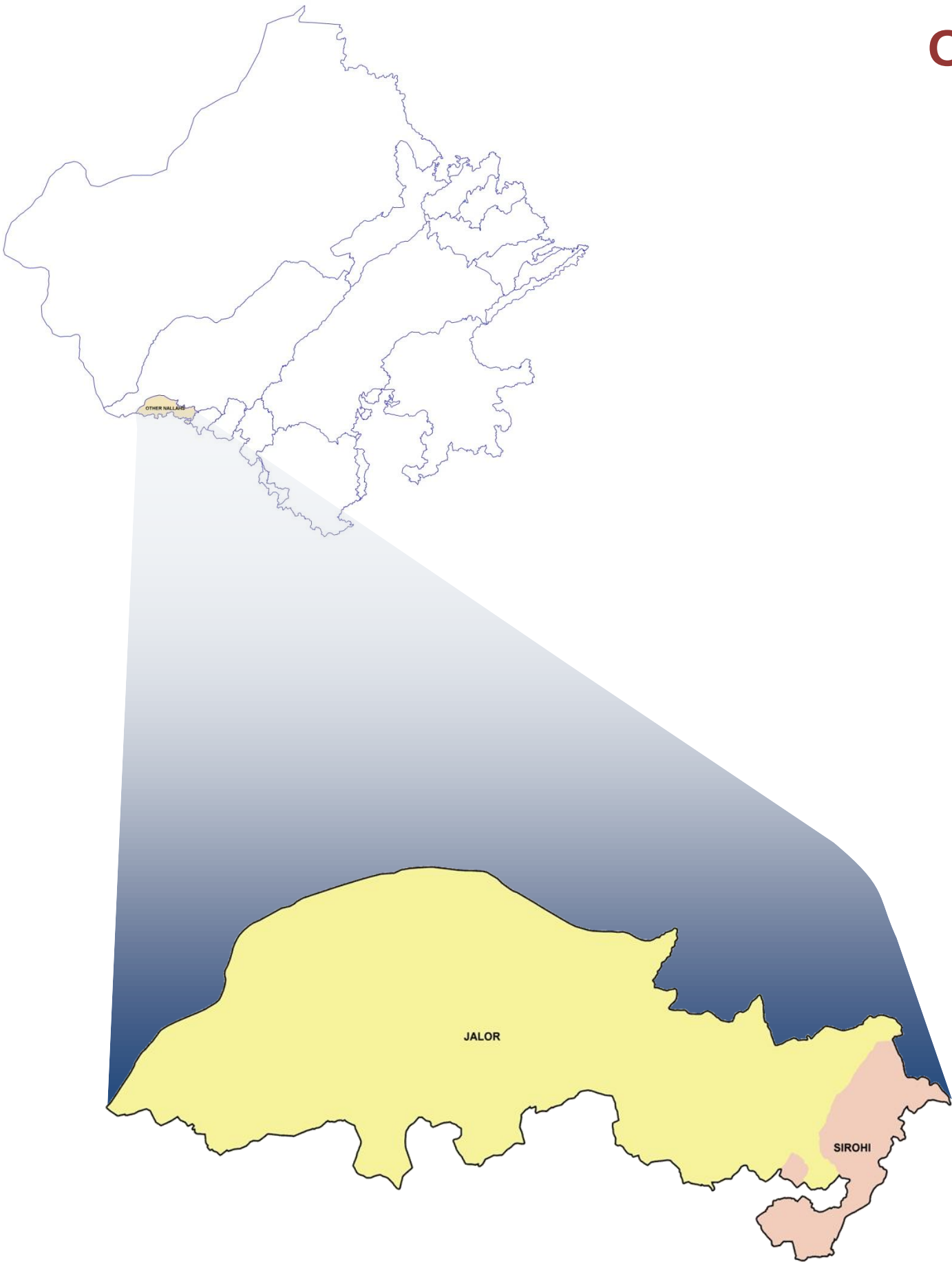
I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

(Arno Schaefer)

Hydrogeological Atlas of Rajasthan

Other Nallahs Basin



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ADMINISTRATIVE SETUP

OTHER NALLAHS BASIN

Location:

This is one of the very small basins in Rajasthan that measure less than 2,000 km² in area (approximately 1900 km²). It is located in southwestern part of the state, bounded in the north and northwest by Luni river basin and in the east by Sukli river basin. The southern border is shared with administrative boundary of Gujarat state. It stretches between 24° 31' 50.08" to 24° 56' 27.96" North latitude and 71° 31' 26.59" to 72° 27' 32.25" East longitudes. Major part of the basin (>90%) falls within Jalor district whereas the remaining area falls in Sirohi district. The basin is a collection of small streams that drain the area but exit the border of Rajasthan State before contributing to any major river. Bandi river, Bargaon, Rip, and Sukal are some of the streams flowing in the area.

The general slope of the terrain is from east to south or west with hills appearing in the east which slowly roll into plains towards west in Sanchore.

Administrative Set-up:

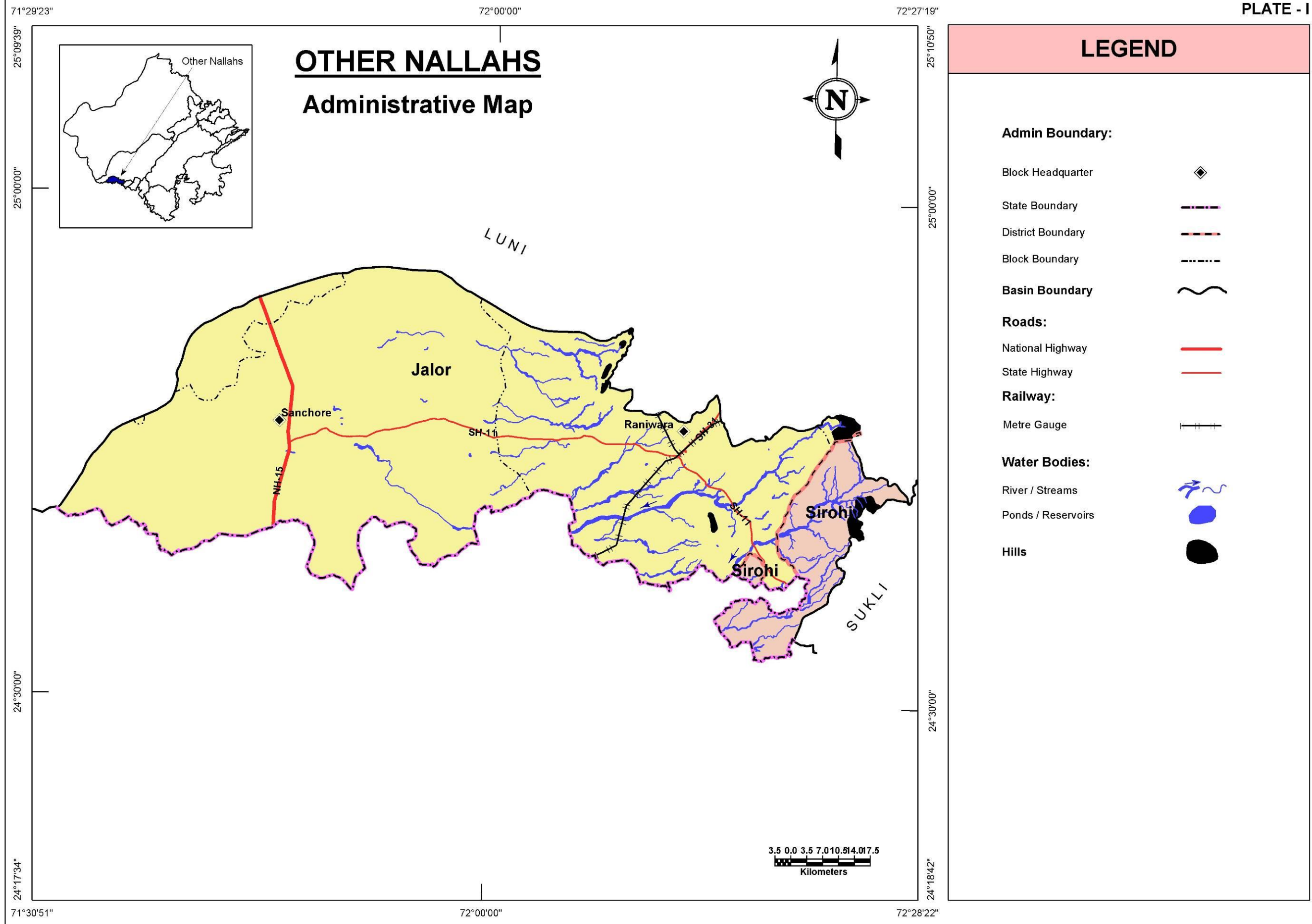
Administratively, Other Nallahs basin extends over parts of Ajmer, Jalor and Sirohi districts which are further divided into 5 blocks, encompassing 154 towns and villages.

S. No.	District Name	Area (sq. km)	% of Basin Area	Total Number of Blocks	Total Number of Towns and Villages
1	Jalor	1,731.2	91.1	4	138
2	Sirohi	168.6	8.9	1	16
Total		1899.8	100.0	5	154

Climate:

The basin falls in Semi-arid to Arid region and experiences extreme temperatures in winter (down to 4°C) and summer (reaches high of 50°C). Rainfall is scanty, with mean annual rainfall of about 353 mm only. As a result the rainy days are very few spread over the months of July to September and for most part of the year it remains dry. Winter ensues rainy season and till February the low temperatures can be felt. Summer sets in from April and May – June months are really hot.

PLATE - I



TOPOGRAPHY

OTHER NALLAHS BASIN

Though the area that forms part of the basin is not large but still the elevation variation is significant, ranging from minimum of 21 m above mean sea level in Jalor district (in the westernmost part) to a maximum of 977m amsl in the eastern hilly parts falling in Sirohi district. The eastern part is hilly and the general slope of the topography is westwards. The hills in Sirohi district and partly in Jalor district are major sources of drainage in the basin which flow westwards.

Table: District wise minimum and maximum elevation

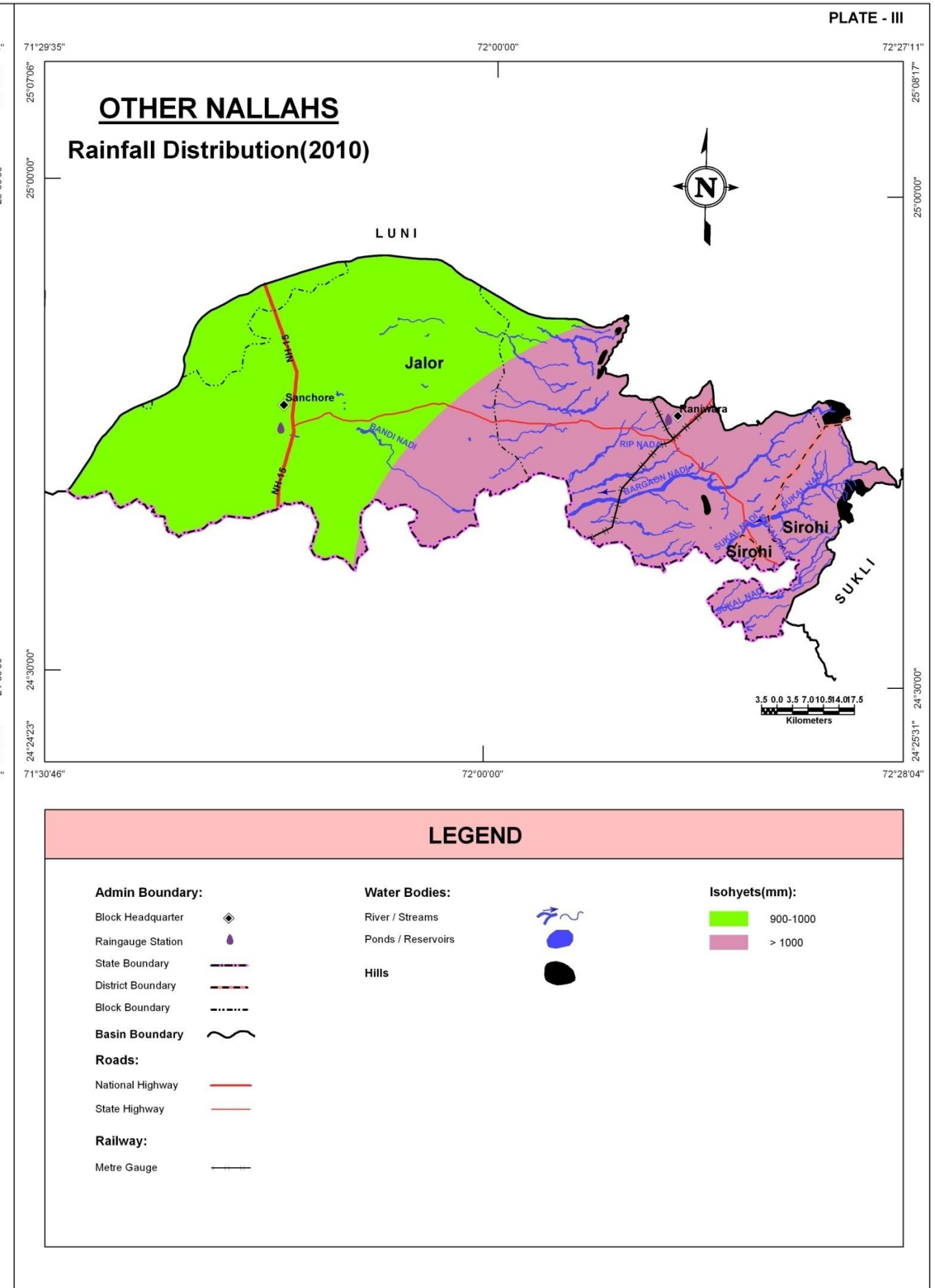
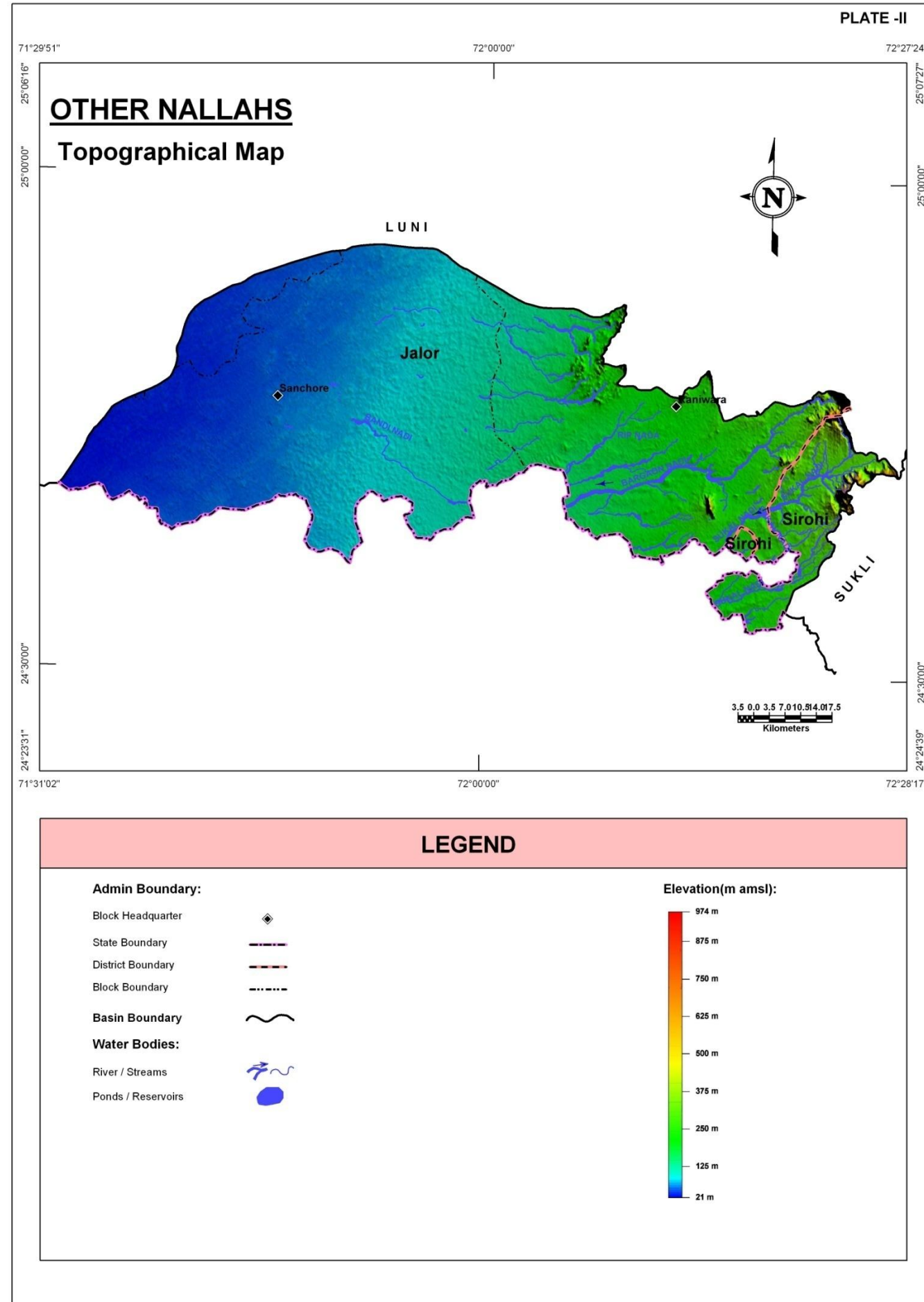
District Name	Min. Elevation (m amsl)	Max. Elevation (m amsl)
Jalor	21.0	850.9
Sirohi	203.7	976.8

RAINFALL

The general distribution of rainfall across the Other Nallah Basin can be visualized from isohyets presented in the Plate III. Total Annual rainfall for the rain gauge stations has been interpolated to understand the distribution across the basin. The year seems to have received a higher rainfall than the other years since it varies from a high of about 1225mm to a low of 938mm indicated by the coloured regions. Higher rainfall is received in the hilly region of eastern part of the basin and reducing gradually towards west.

Table: District wise total annual rainfall (based on year 2010 meteorological station recordings (<http://waterresources.rajasthan.gov.in>))

S. No.	Rain gauge Stations	Total Monsoon Rainfall (mm)	Total Non-Monsoon Rainfall (mm)	Total Annual Rainfall (mm)
1	Raniwara	1,064.0	161.0	1,225.0
2	Sanchoe	820.0	118.0	938.0



GEOLOGY

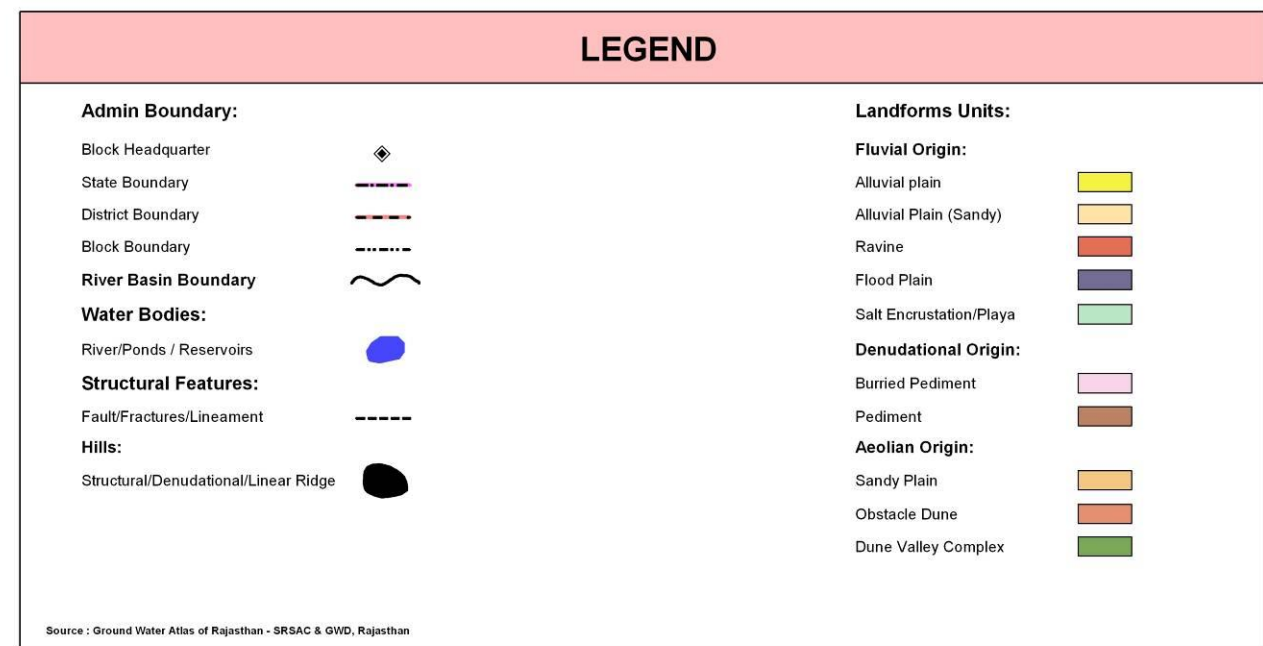
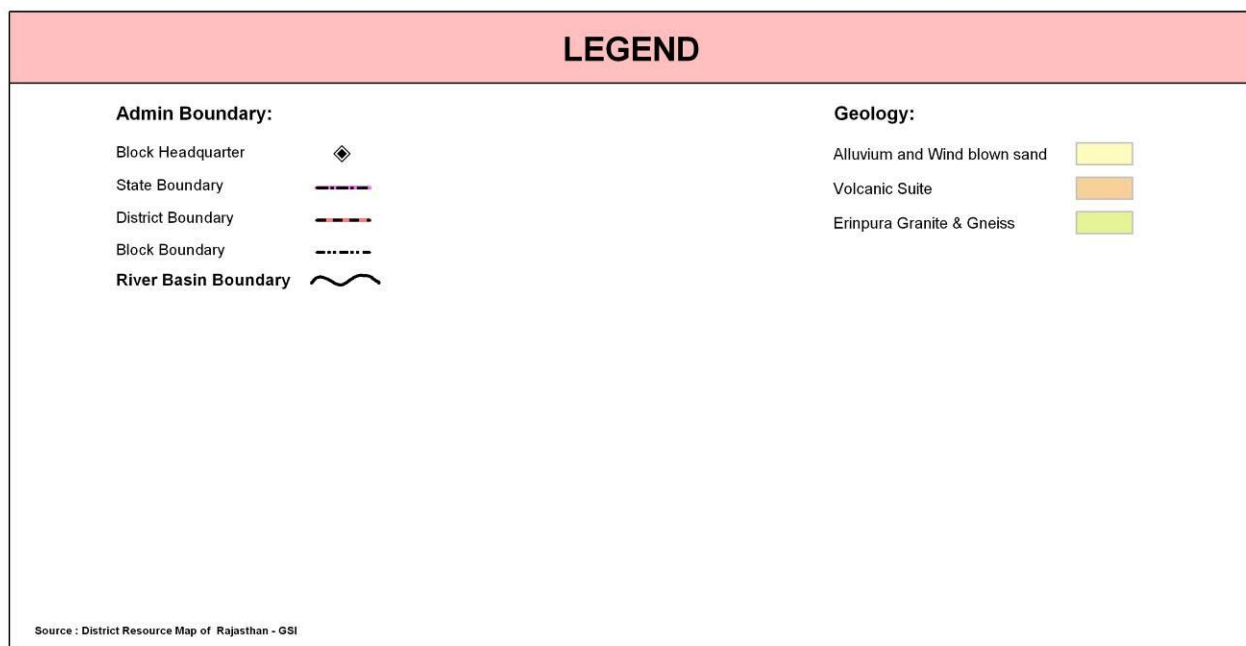
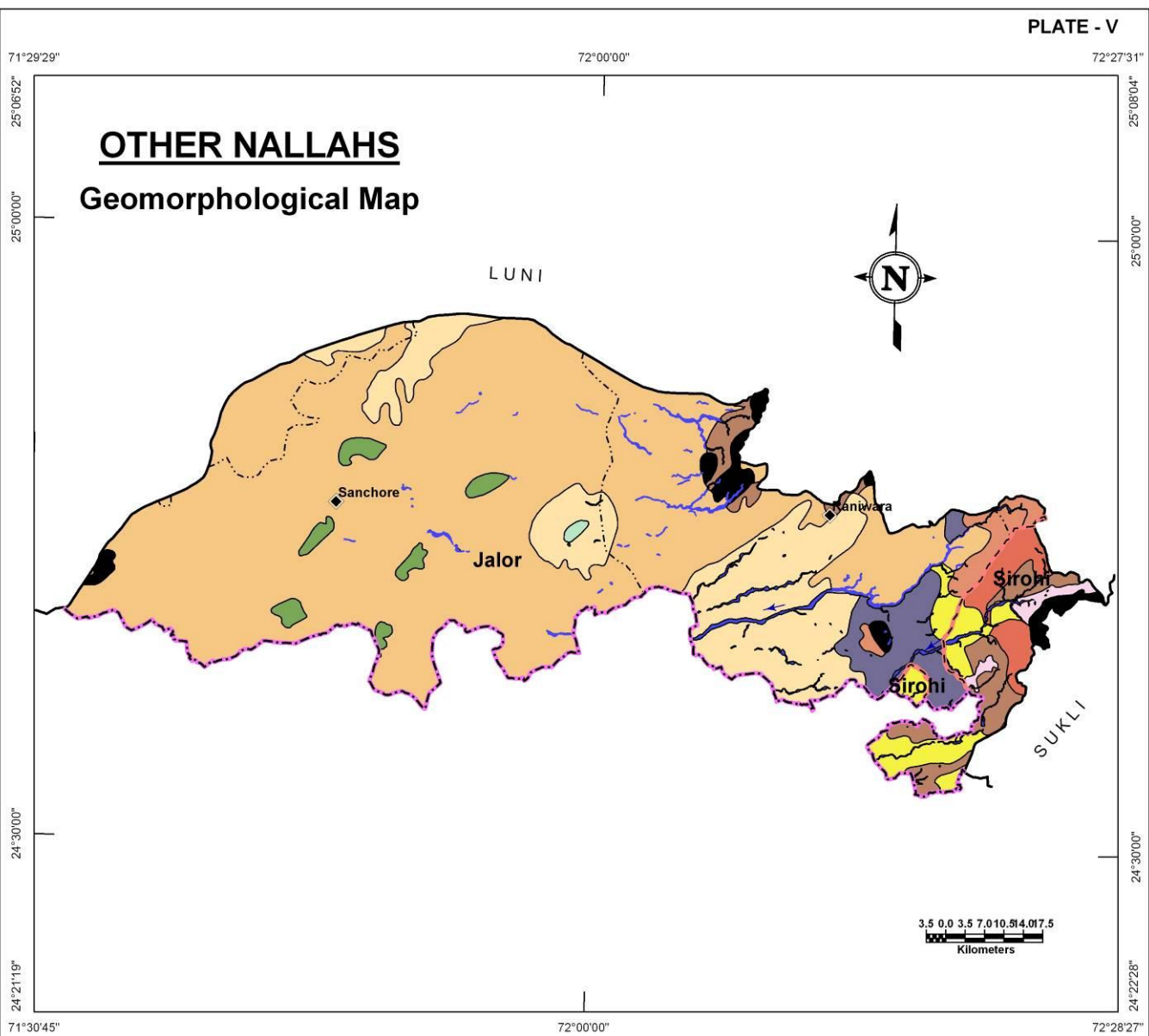
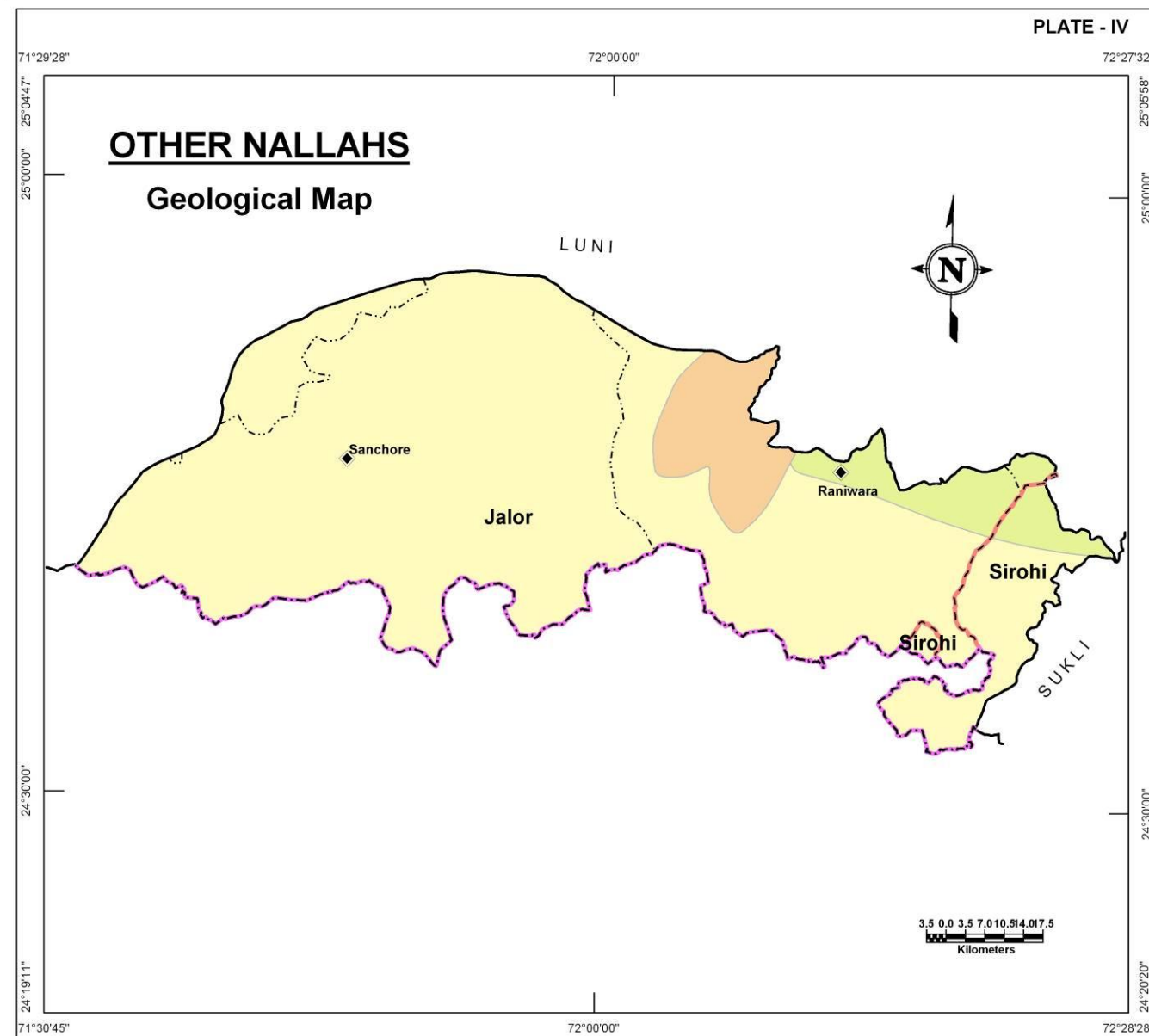
OTHER NALLAHS BASIN

The hills are largely formed by Delhi Super Group rocks whereas the alluvial cover in the western part of the basin is very pronounced. The sand cover of Recent to Sub-Recent age is primarily windblown sand with minor contribution from alluvial sources.

Age	Super-Group	Group/ Formation	Rock Types
Sub-Recent to Recent	Aeolian Deposits (and Colluvium)	Alluvium	Windblown sand (sand dunes) cobble, pebble, gravel, sand, silt, clay and kankar
-----x-----x-----x-----Unconformity-----x-----x-----x-----			
Lower Precambrian to Upper Precambrian	Lower Vindhyan	Malani Rhyolite	Rhyolitic lavas with abundant pyroclastic material
	Post-Delhi (extrusives and intrusives)	Erinpura (Sandra Ambaji) granite and gneiss	Granites, gneisses, pegmatites, Amphibolites, epidiorites
Middle Proterozoic	Delhi	Sirohi	Phyllites, mica schists, Dolomites, marbles and gneisses

GEOMORPHOLOGY

Origin	Landform Unit	Description
Aeolian	Dune Valley Complex	Cluster of dunes and interdunal spaces with undulating topography formed due to wind-blown activity, comprising of unconsolidated sand and silt.
	Obstacle Dune	Formed on windward/leeward sides of obstacle like isolated hills or continuous chain of hills, dune to obstruction in path of sand laden winds. Badly dissected well cemented and vegetated.
	Sandy Plain	Formed of aeolian activity, wind-blown sand with gentle sloping to undulating plain, comprising of coarse sand, fine sand, silt and clay.
Denudational	Buried Pediment	Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.
	Pediment	Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.
Fluvial	Alluvial Plain	Mainly undulating landscape formed due to fluvial activity, comprising of gravels, sand, silt and clay. Terrain mainly undulating, produced by extensive deposition of alluvium.
	Alluvial Plain (Sandy)	Flat to gentle undulating plain formed due to fluvial activity, mainly consists of gravels, sand, silt and clay with unconsolidated material of varying lithology, predominantly sand along river.
	Flood Plain	The surface or strip of relatively smooth land adjacent to a river channel formed by river and covered with water when river over flows its bank. Normally subject to periodic flooding.
	Ravine	Small, narrow, deep, depression, smaller than gorges, larger than gulley, usually carved by running water.
	Salt Encrustation/Playa	Topographical depression comprising of clay, silt, sand and soluble salts, usually undrained and devoid of vegetation.
Hills		Hills and intervening valleys in hilly areas



AQUIFERS

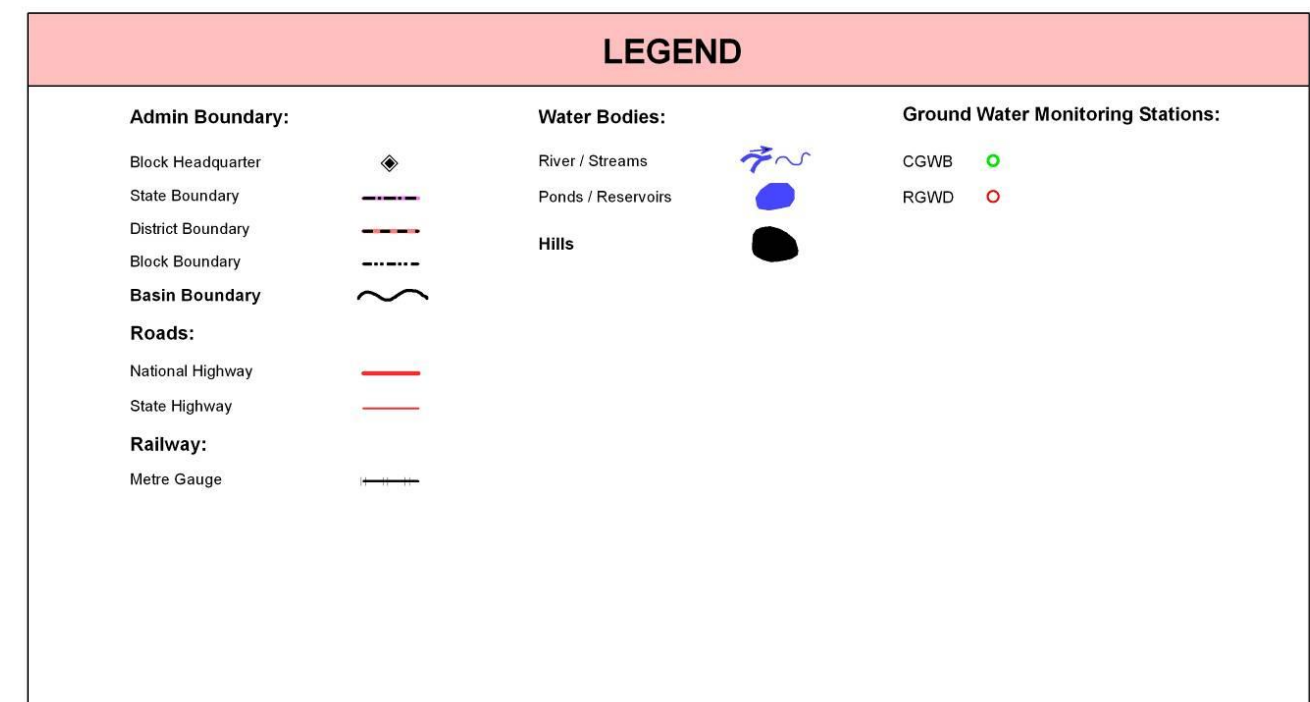
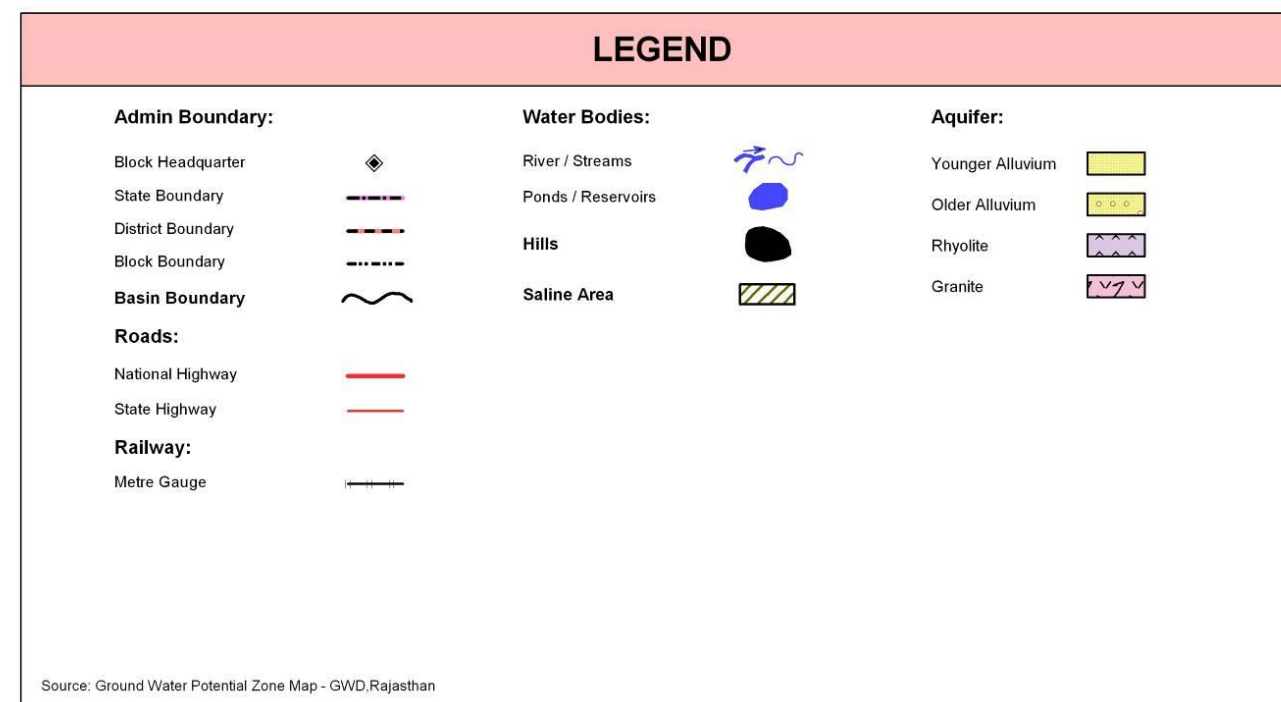
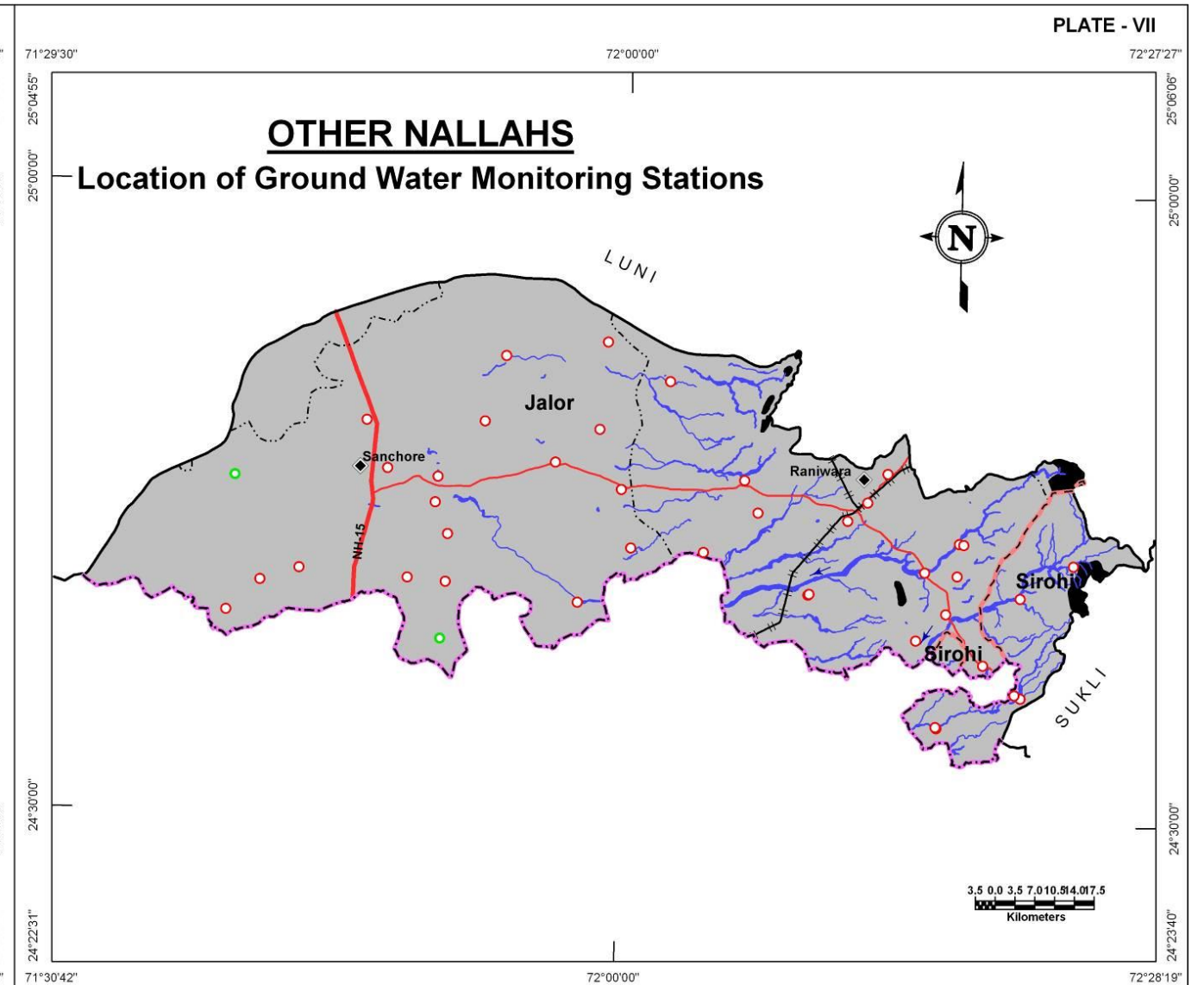
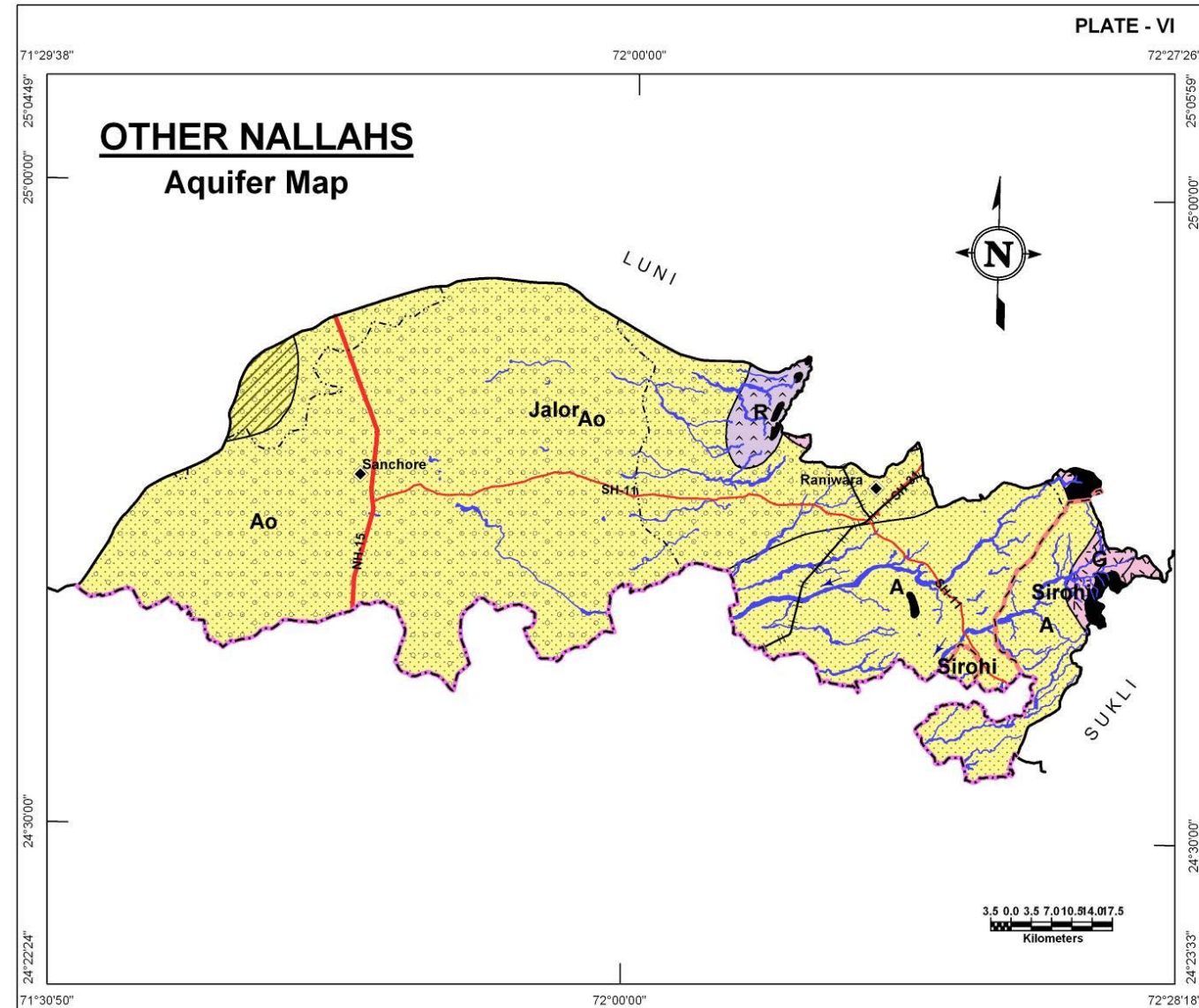
While aquifers are formed in weathered and fractured hardrocks of Rhyolite and Granite, the alluvial aquifers formed in Older and younger alluviums are primary sources of groundwater in the basin. The alluvial aquifers together contribute to 96% of the sub surface distribution whereas the hardrock aquifers just make about 3%. The rhyolite aquifer is seen to the west of Raniwara in Jalor district around the hilly area, the granitic aquifers are seen in the easternmost part of the basin in Sirohi district adjacent to hills. The alluvial aquifers occupy the whole of rest of the basin.

Aquifer in Potential Zone	Area (sq km)	% of Basin Area	Description of the unit/Occurrence
Younger Alluvium	454.9	23.9	It is largely constituted of aeolian and fluvial sand, silt, clay, gravel and pebbles in varying proportions.
Older Alluvium	1,363.3	71.8	This litho-unit comprises of mixture of heterogeneous fine to medium grained sand, silt and kankar
Rhyolite	37.2	2	Rhyolite is porphyritic and has phenocryst of quartz and feldspar.
Granite	28.3	1.5	Light grey to pink colour, medium to coarse grained, and characteristically have porphyritic texture
Non Potential Zone	16.1	0.8	Hills and reserve forests
TOTAL	1,899.8	100.0	

LOCATION OF GROUND WATER MONITORING WELLS

The basin has a network of 42 ground water monitoring stations (40) in the basin owned by RGWD and CGWB (2). To further strengthen the network in terms of water quality and water level monitoring, a total of 19 wells have been recommended for Jalor (18 wells) and Sirohi (1) districts.

District Name	Existing Ground Water Monitoring Stations			Recommended Additional Ground Water Monitoring Stations	
	CGWB	RGWD	Total	Water Level	Water quality
Jalor	2	34	36	0	18
Sirohi		6	6	0	1
Total	2	40	42	0	19



LOCATION OF EXPLORATORY WELLS

In all there are 41 exploratory wells present in the basin drilled in the past by RGWD (37) and CGWB (4) that form basis for understanding of sub-surface and aquifer distribution.

District	Exploratory Wells		
	CGWB	RGWD	Total
Jalor	3	31	34
Sirohi	1	6	7
Total	4	37	41

OTHER NALLAHS BASIN

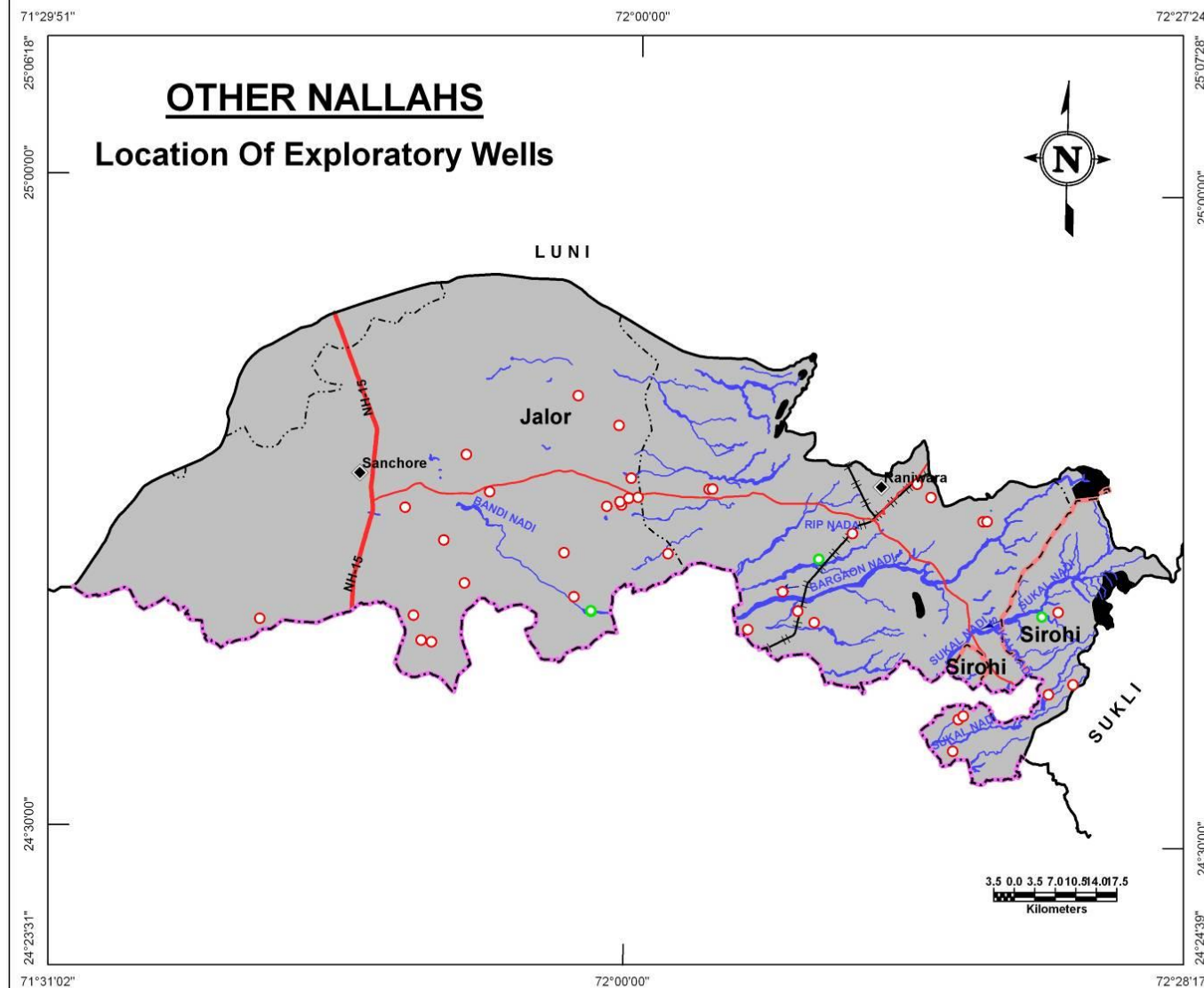
DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

The general depth to water level in the basin ranges from 10 to 40 meters below ground level, as seen in the eastern and western parts of the basin. The central part of the basin however, shows higher depth to water levels reaching upto 70m bgl.

Depth to water level (m bgl)	District wise area coverage (sq km)*		Total Area (sq km)
	Jalor	Sirohi	
< 10	247.1	0.7	247.8
10 - 20	364.2	72.5	436.7
20 - 30	408.1	81.9	490.0
30 - 40	332.2	6.5	338.7
40 - 50	233.7	-	233.7
50 - 60	114.4	-	114.4
60 - 70	22.3	-	22.3
> 70	0.1	-	0.1
Total	1,722.1	161.6	1,883.7

* The area covered in the derived maps is less than the total basin area since the hills have been excluded from interpolation/contouring.

PLATE -VIII



LEGEND

Admin Boundary:

- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Roads:**
- National Highway
- State Highway
- Railway:**
- Metre Gauge

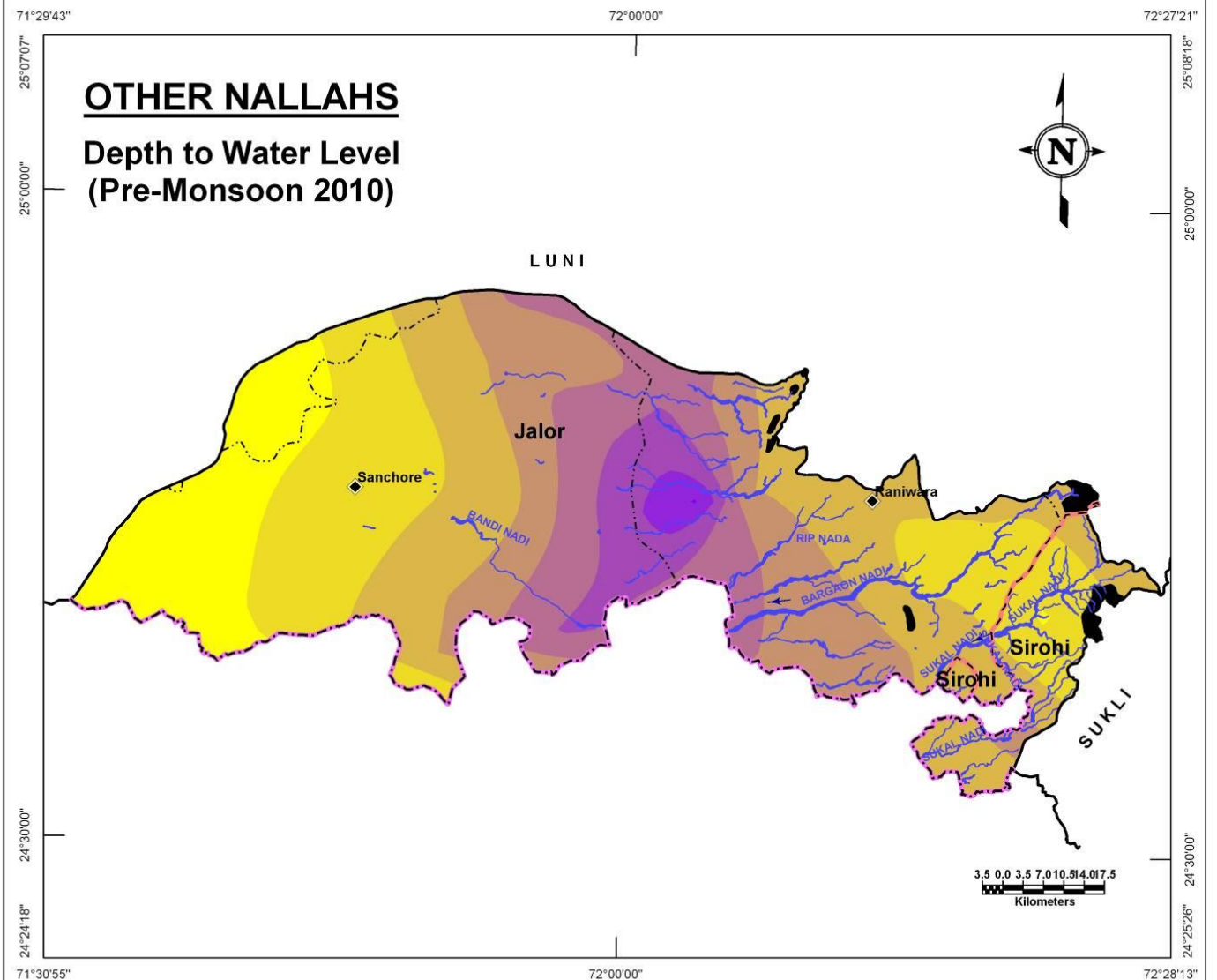
Water Bodies:

- River / Streams
- Ponds / Reservoirs
- Hills

Exploratory Wells:

- CGWB
- RGWD

PLATE - IX



LEGEND

Admin Boundary:

- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:**
- River / Streams
- Ponds / Reservoirs
- Hills

Depth to Water Level(m bgl):

- < 10
- 10 - 20
- 20 - 30
- 30 - 40
- 40 - 50
- 50 - 60
- 60 - 70
- > 70

WATER TABLE ELEVATION (PRE MONSOON 2010)

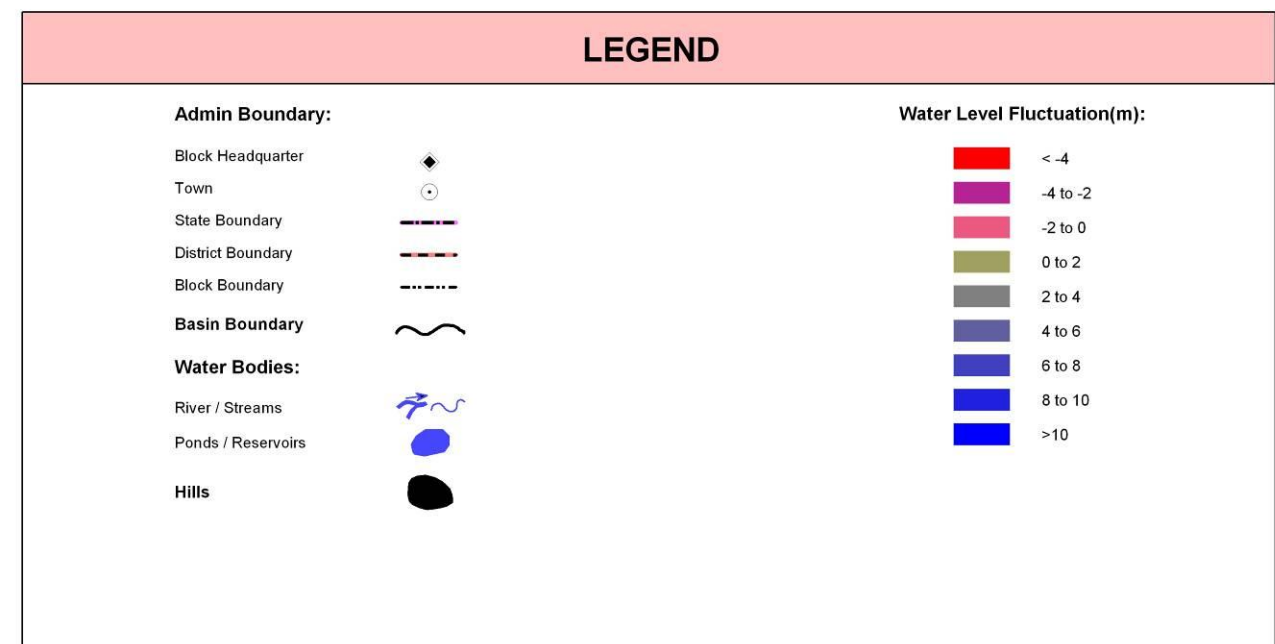
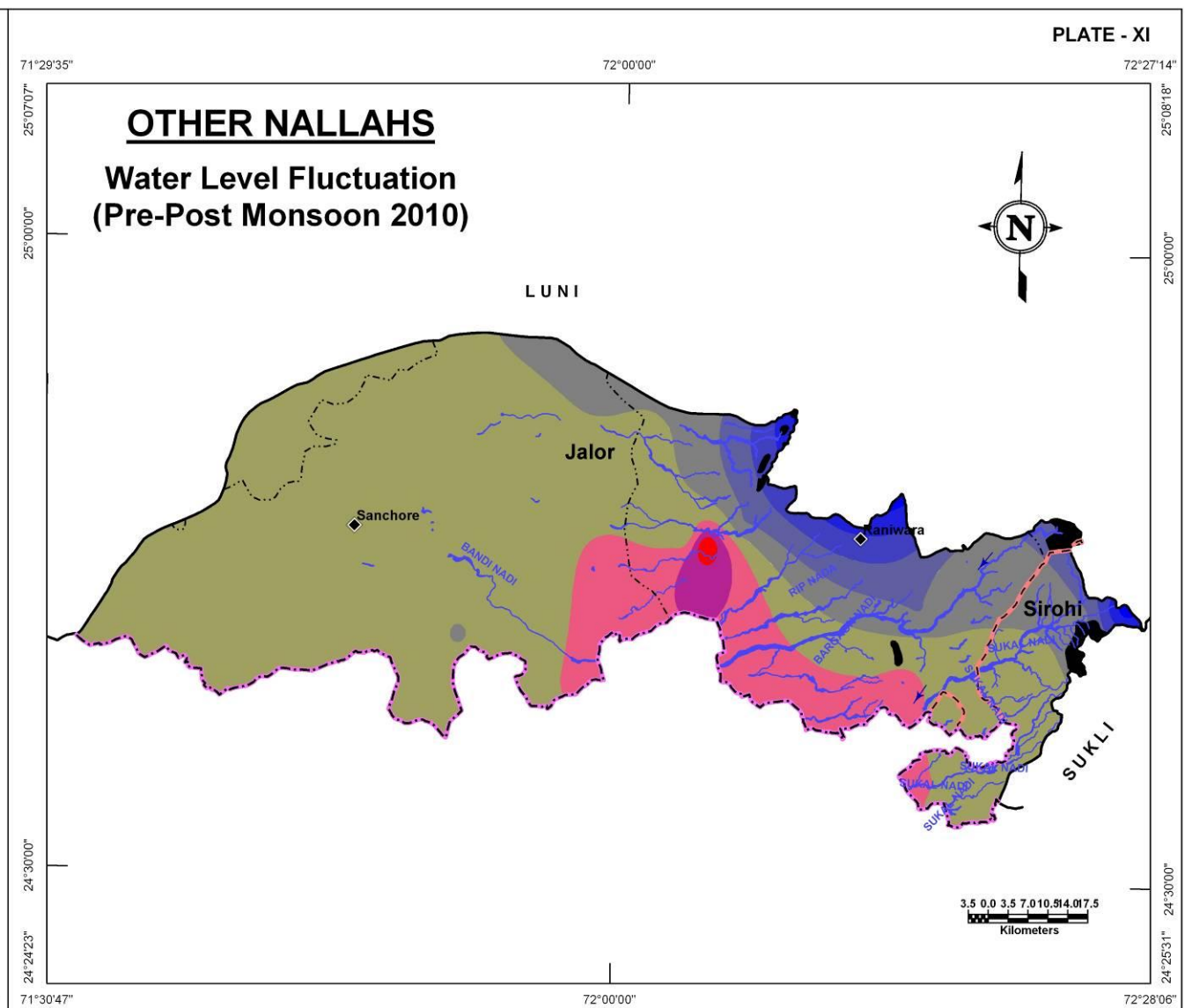
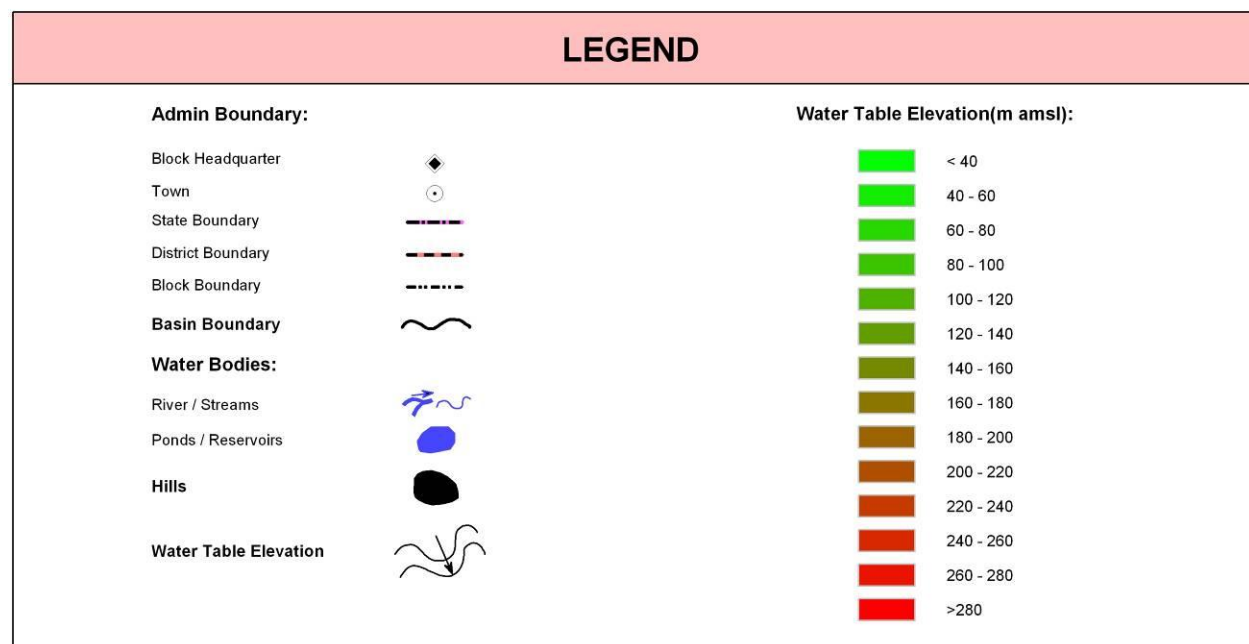
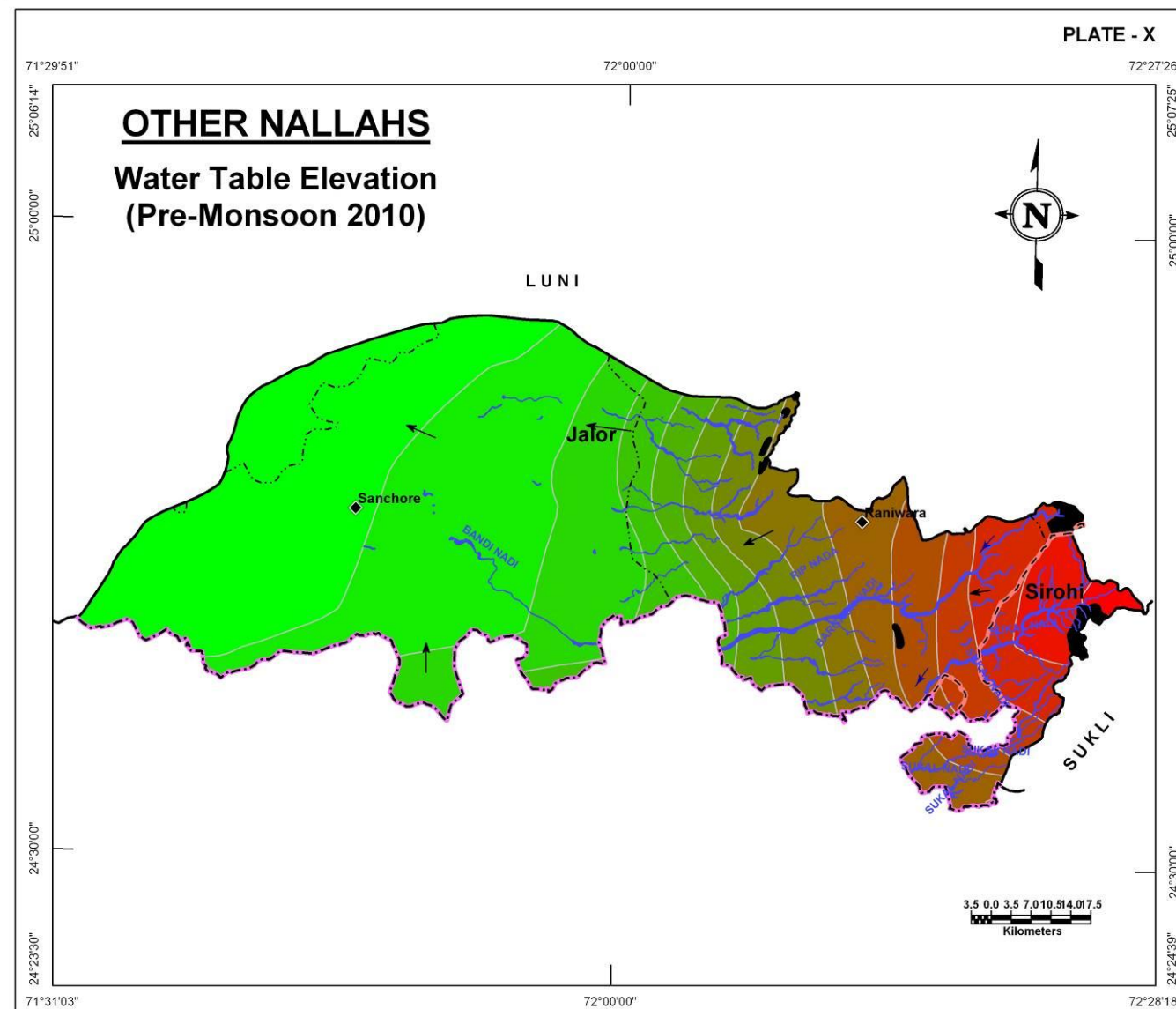
Water table elevation shows variation from less than 40m above mean sea level to about 280m amsl. On comparing this map with topography map it becomes evident that high water table corresponds to hilly areas in the east and has steeper gradients when moved to west. In the western parts of thick alluvial aquifers, the water table has lower elevations and also shows a flattening of the gradient with sluggish flows.

Depth to water table (m amsl)	District wise area coverage (sq km)		Total Area (sq km)
	Jalor	Sirohi	
< 40	502.5	-	502.5
40 - 60	395.2	-	395.2
60 - 80	201.3	-	201.3
80 - 100	74.4	-	74.4
100 - 120	58.6	-	58.6
120 - 140	76.8	-	76.8
140 - 160	89.5	-	89.5
160 - 180	93.0	-	93.0
180 - 200	74.7	26.8	101.5
200 - 220	59.4	17.5	76.9
220 - 240	49.0	13.8	62.8
240 - 260	47.7	42.8	90.5
260 - 280	-	50.9	50.9
> 280	-	9.8	9.8
Total	1,722.1	161.6	1,883.7

WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

The hardrock aquifer/hilly areas show higher fluctuation in ground water levels indicating rise of 10m from pre monsoon water levels. In the alluvial aquifer/plain areas, which occupy most part of the basin, the water level fluctuation has been in the range of +2 to -2 m. There is an unusual fall of water level by around 4 meters in the central part of the area which appears to be due to some localized high ground water extraction.

District Name	District wise area coverage (sq km) within fluctuation range (m)									Total Area (sq km)
	< -4	-4 to -2	-2 to 0	0 to 2	2 to 4	4 to 6	6 to 8	8 to 10	> 10	
Jalor	3.4	25.6	202.9	1,172.7	188.2	77.5	37.8	12.7	1.3	1,722.1
Sirohi	-	-	8.0	102.5	33.8	10.6	2.8	3.0	0.9	161.6
Total	3.4	25.6	210.9	1,275.2	222.0	88.1	40.6	15.7	2.2	1,883.7



ELECTRICAL CONDUCTIVITY DISTRIBUTION

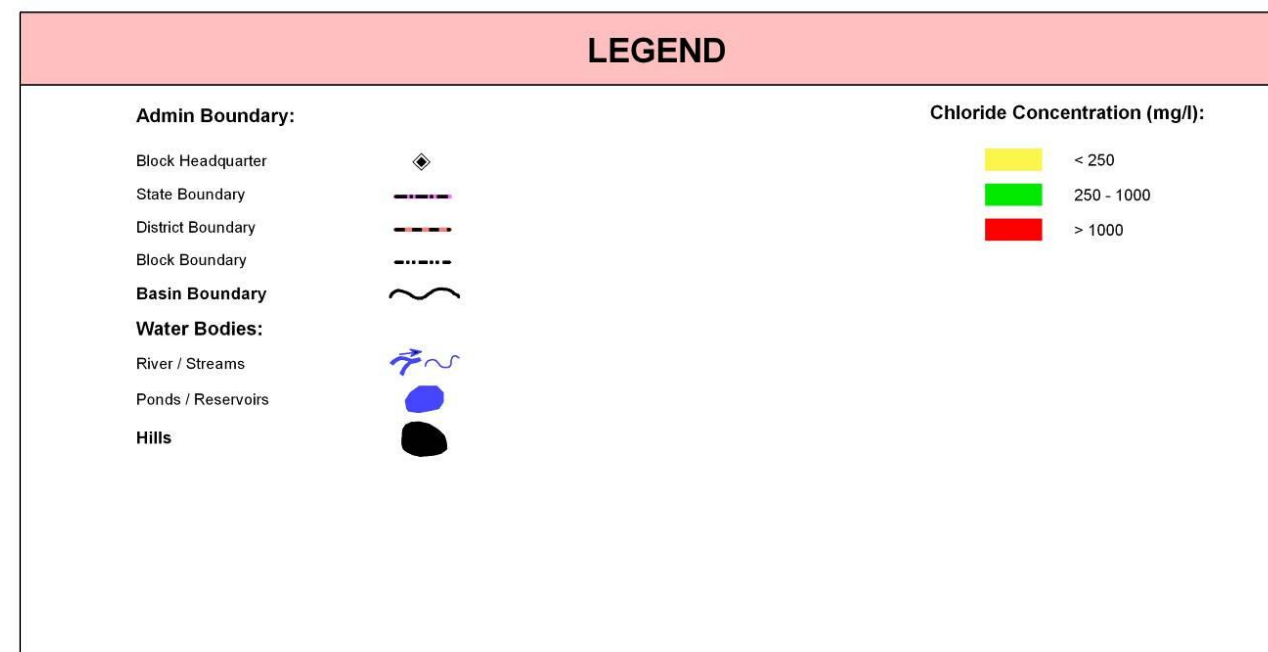
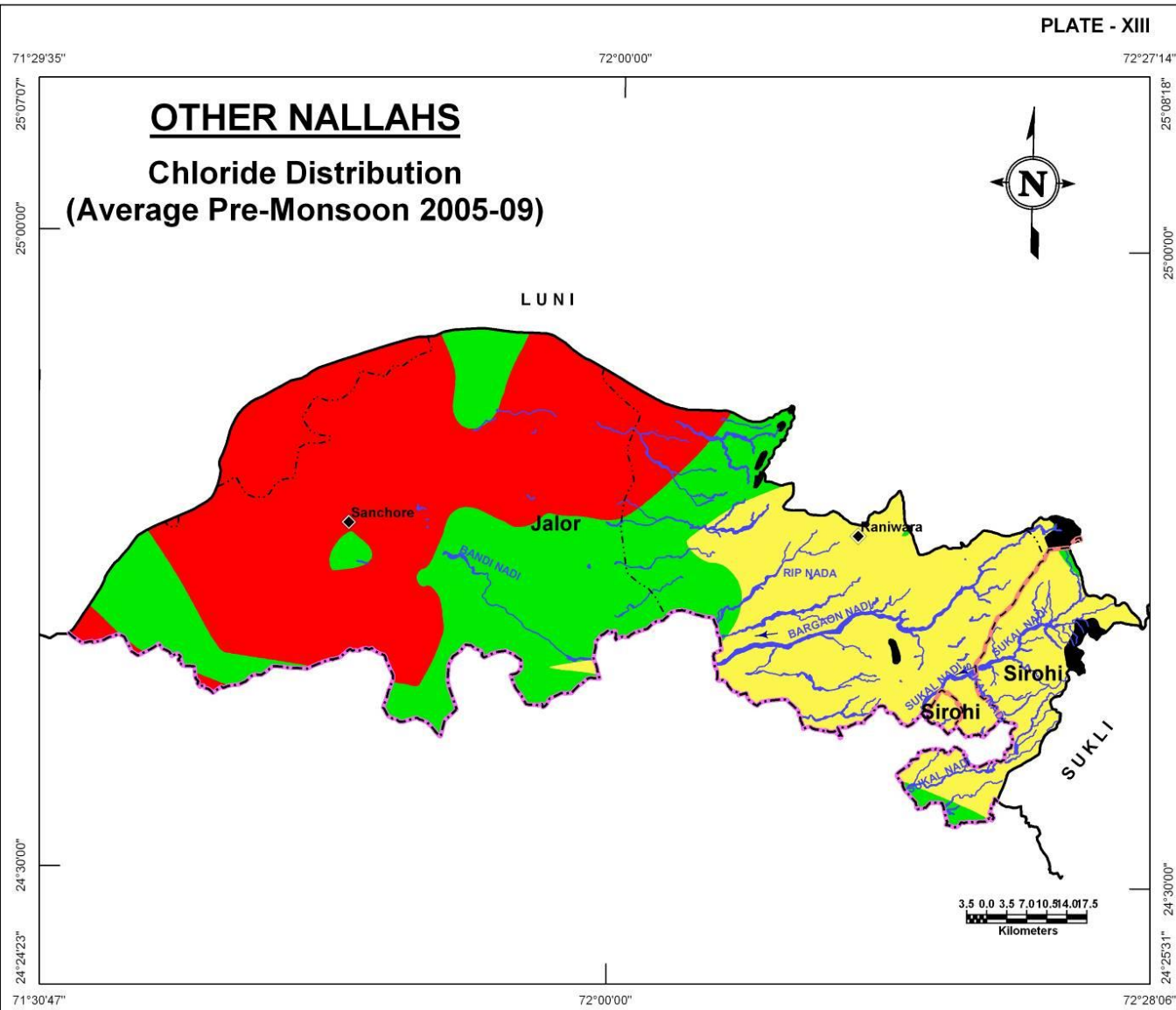
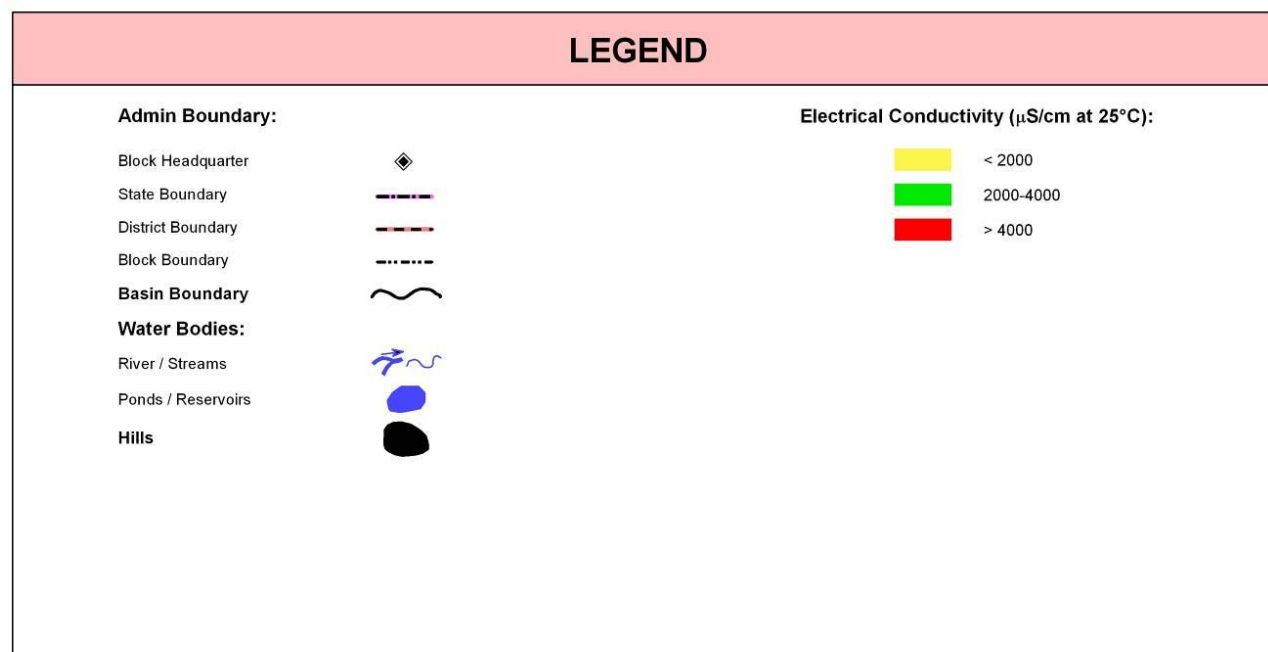
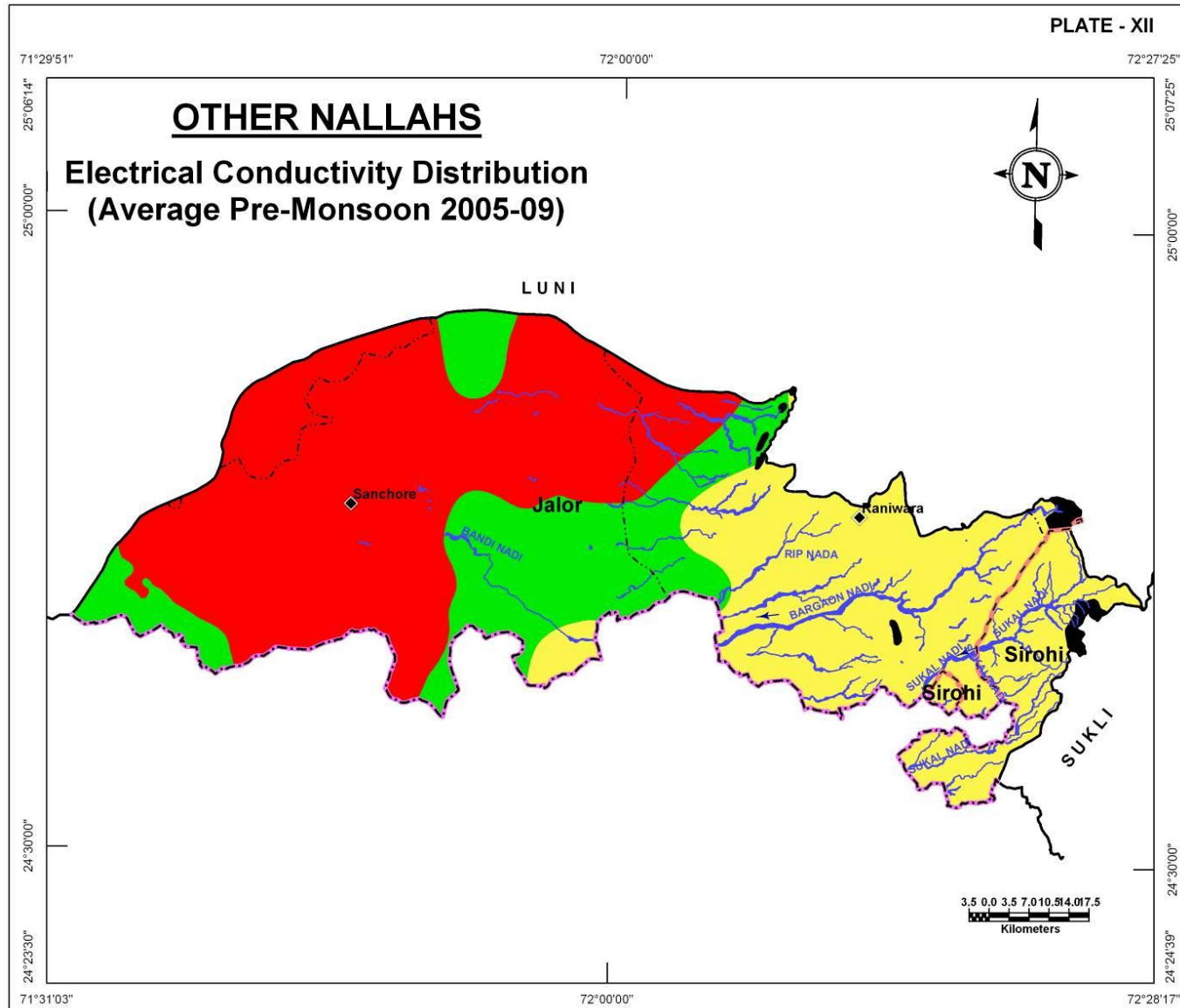
The Electrical Conductivity (at 25°C) distribution map is presented in Plate – XII shows a significantly large patch of red colour (EC >4000 $\mu\text{S}/\text{cm}$) in the western alluvial part of the basin occupying about 46% of the area. The areas in the eastern half of the basin however, show the presence of good ground water quality having EC < 2000 $\mu\text{S}/\text{cm}$ (represented by Yellow coloured regions) which occupies approximately 33% of the basin area and is suitable for all purposes. The remaining area (about 20% of the basin) has moderate EC value (2000-4000 $\mu\text{S}/\text{cm}$).

Electrical Conductivity Ranges (μS/cm at 25°C) (Ave. of years 2005-09)	District wise area coverage (sq km)				Total Area (sq km)
	Jalor		Sirohi		
	Area	% age	Area	% age	
< 2000	470.6	27.3	161.6	100.0	632.2
2000-4000	385.0	22.4	-	-	385.0
> 4000	866.5	50.3	-	-	866.5
Total	1,722.1	100.0	161.6	100.0	1,883.7

CHLORIDE DISTRIBUTION

High chloride concentration in ground water may also render it unsuitable for domestic and other purposes. The red coloured regions in Plate – XIII are such areas where Chloride concentration is very high (> 1000 mg/l) which are largely found in alluvial aquifers in the western part of the basin. Good quality water with low Chloride concentration (<250 mg/l) are found in the close proximity of hilly areas as seen in the eastern part of the basin where the water table gradient has also been noticed to be higher than the western part. About 25% of the basin area located in central part of the basin between the high and low Chloride value areas, has moderate Chloride concentration in ground water that ranges in between 250-1000 mg/l.

Chloride Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)				Total Area (sq km)
	Jalor		Sirohi		
	Area	% age	Area	% age	
< 250	425.9	24.7	150.7	93.2	576.6
250 - 1000	508.5	29.6	10.9	6.8	519.4
> 1000	787.7	45.7	-	-	787.7
Total	1,722.1	100.0	161.6	100.0	1,883.7



FLUORIDE DISTRIBUTION

The Fluoride concentration map (Plate XIV) displays a number of scattered patches of high fluoride concentration (>3 mg/l) in the west, north and eastern parts of the basin of which the one around Sanchore is the largest one occupying nearly 4% of the area whereas the rest of four patches together constitute the rest 6% of the high fluoride concentration area. The rest of the area has moderate to low concentration of fluoride in ground water.

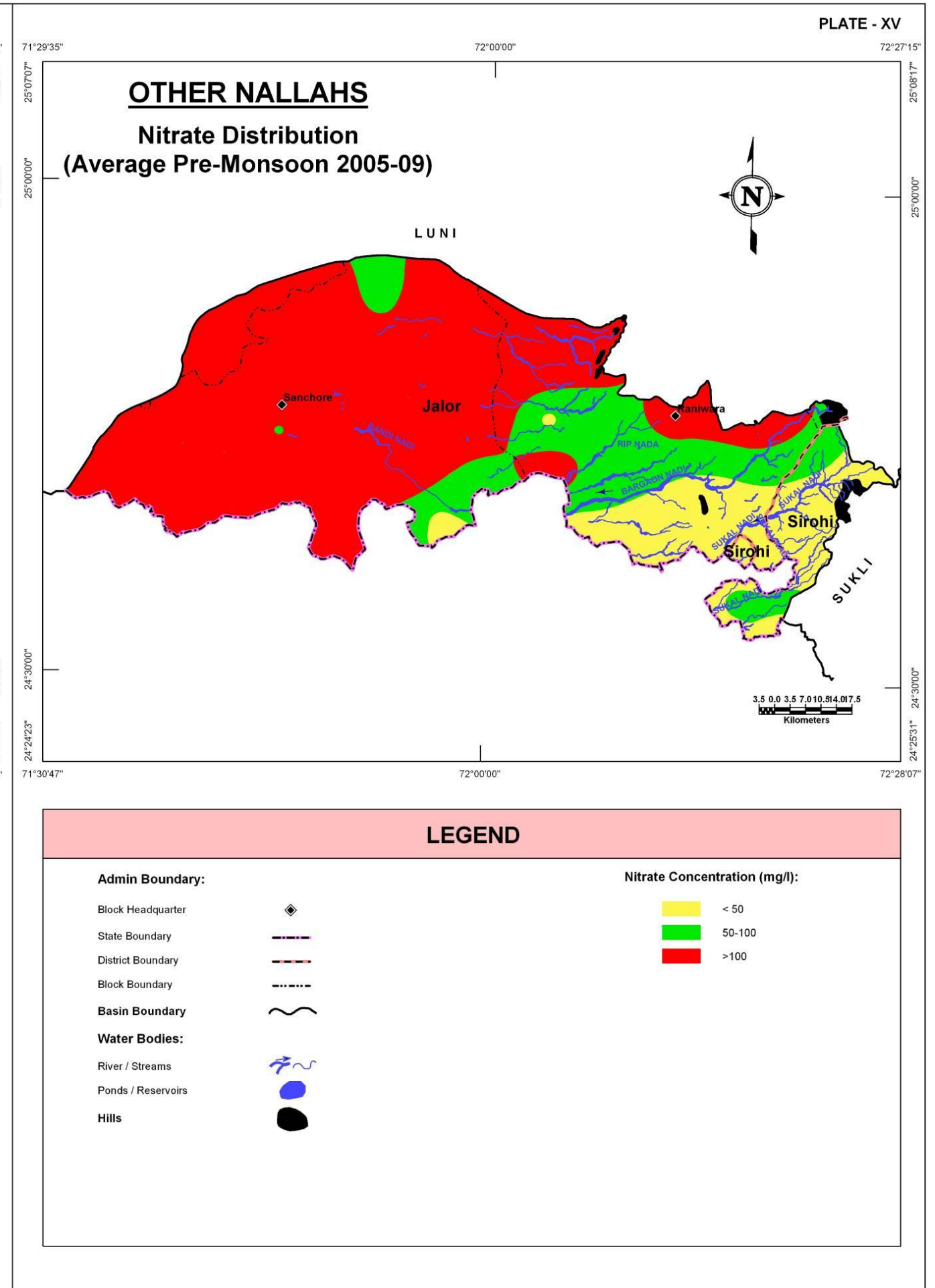
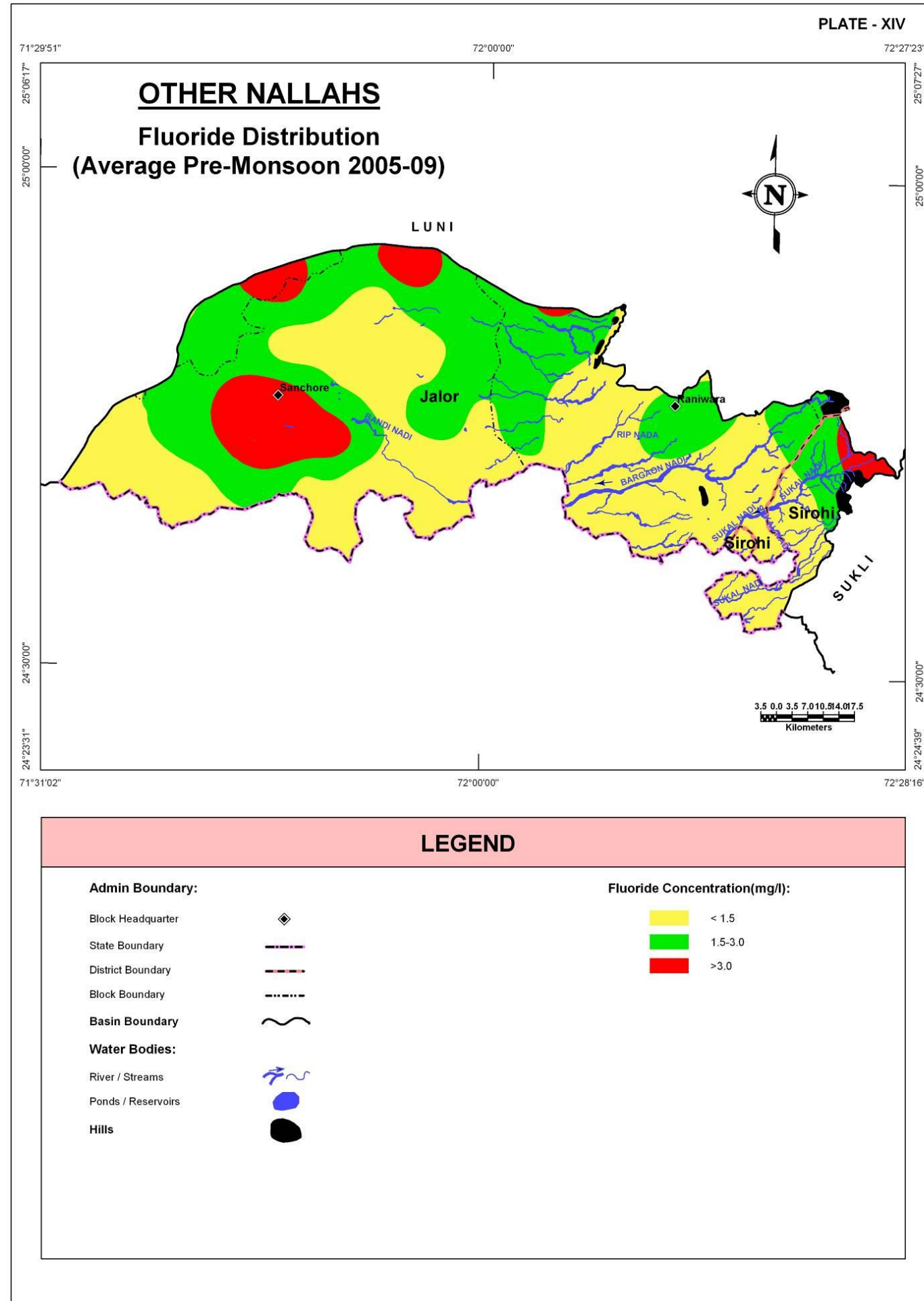
Fluoride Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)				Total Area (sq km)
	Jalor		Sirohi		
	Area	% age	Area	% age	
< 1.5	857.6	49.8	103.3	63.9	960.9
1.5-3.0	694.3	40.3	42.3	26.2	736.6
> 3.0	170.2	9.9	16.0	9.9	186.2
Total	1,722.1	100.0	161.6	100.0	1,883.7

OTHER NALLAHS BASIN

NITRATE DISTRIBUTION

Information of Nitrate concentration helps in understanding the suitability of ground water for agricultural purposes. A perusal of Plate – XV reveals a significantly large patch accounting for more than 65% of the basin area in the alluvial plains falls under high Nitrate category (>100 mg/l). Another 20% of the alluvial area shows presence of moderately high Nitrate concentration in ground water (i.e., 50 – 100 mg/l) which too is not very much suitable for agriculture but some crops may be grown. Hardly, 15% of the basin area (which also includes non-alluvial areas, contains low (<50 mg/l) Nitrate in the southeastern parts of the basin.

Nitrate Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)				Total Area (sq km)
	Jalor		Sirohi		
	Area	% age	Area	% age	
< 50	172.1	10.0	119.0	73.6	291.1
50-100	325.6	18.9	42.6	26.4	368.2
> 100	1,224.4	71.1	-	-	1,224.4
Total	1,722.1	100.0	161.6	100.0	1,883.7



DEPTH TO BEDROCK

Plate – XVI represents distribution of bedrock in terms of depth to bedrock map as meters below ground level. A perusal of map reveals that in general, the depth to bedrock in the basin is shallow in the east where depth ranges between negligible to 60m below ground level, moderately deep in the western part where the depth ranges between 60 to 100m bgl. In the central part of the basin the deepest depth is encountered which is about 120m bgl.

Depth to Bedrock (m bgl)	District wise area coverage (sq km)		Total Area (sq km)
	Jalor	Sirohi	
< 40	-	31.4	31.4
40-60	307.7	130.2	437.9
60-80	42.0	-	42.0
80-100	786.3	-	786.3
100-120	497.6	-	497.6
> 120	88.5	-	88.5
Total	1,722.1	161.6	1,883.7

UNCONFINED AQUIFER

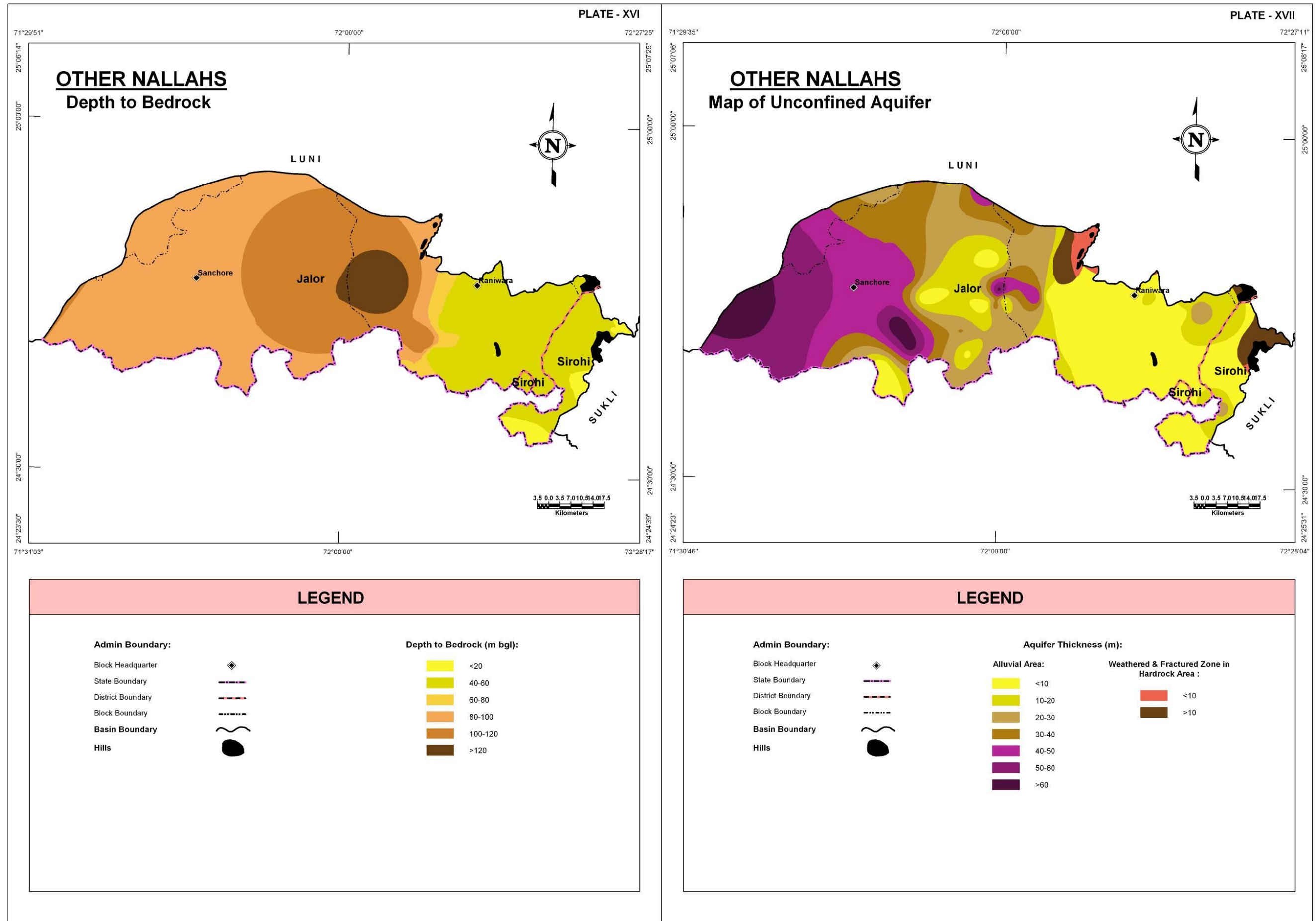
Hydrogeological properties are different for alluvial and hard rock aquifers and therefore, this aquifer has been mapped as two separate regions viz, unconfined aquifers in alluvial and in hard rock areas. Aquifers formed in weathered/fractured zone of hard rocks occupy very limited areas i.e., only about 65 sq km (less than 4% of basin area) whereas the rest of the basin has aquifers formed in alluvial material which attain high thicknesses of more than 40m reaching about 60m in western part of the basin. In the eastern part of the basin the thickness of alluvial aquifers is however low, estimated to be less than 20m.

Alluvial areas:

Unconfined aquifer Thickness (m)	District wise area coverage (sq km)		Total Area (sq km)
	Jalor	Sirohi	
< 10	403.4	104.3	507.7
10-20	217.6	29.6	247.2
20-30	283.4	3.1	286.5
30-40	218.9	-	218.9
40-50	260.0	-	260.0
50-60	227.0	-	227.0
> 60	71.0	-	71.0
Total	1,681.3	137.0	1,818.3

Hardrock areas:

Unconfined aquifer Thickness (m)	District wise area coverage (sq km)		Total Area (sq km)
	Jalor	Sirohi	
< 10	17.4	1.1	18.5
> 10	23.4	23.5	46.9
Total	40.8	24.6	65.4



CROSS SECTIONS

OTHER NALLAHS BASIN

Several hydrogeologic cross sections have been drawn to better depict the sub-surface distribution of lithology. These sections have been overlaid with geological maps and structural faults if there are any have been transferred for verification of their impact on sub-surface material disposition. The alignment of the cross sections is shown in Plate – XVIII and corresponding sections are presented in Plates – XIX to XXIII. The general alignment of the sections is as given below:

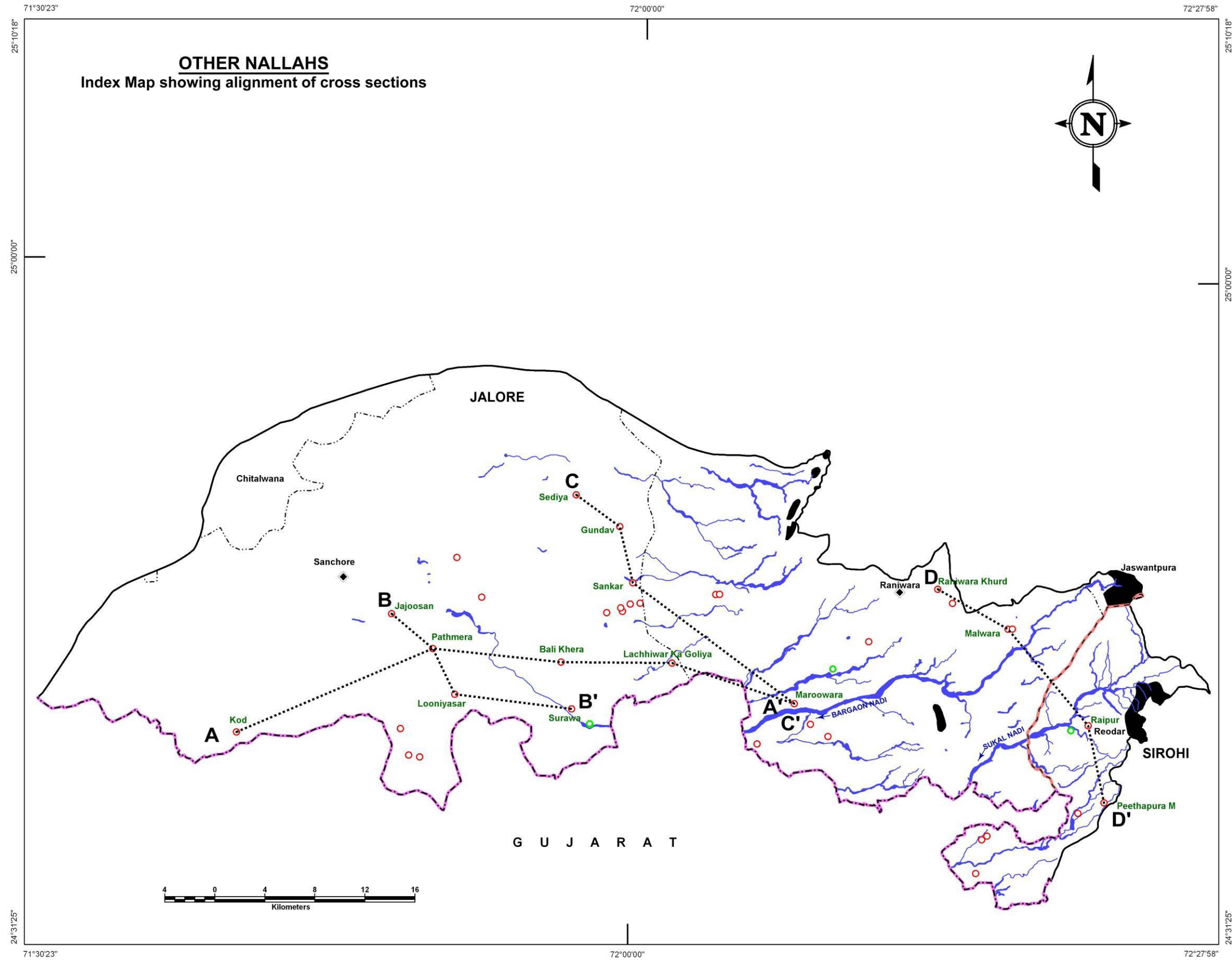
Name of Section Line	Orientation
Section AA'	W–E
Section BB'	NW–SE
Section CC'	NW–SE
Section DD'	NW–SE

PLATE - XVIII

OTHER NALLAHS
Index Map showing alignment of cross sections



- Legend**
- ◆ Block Headquarter
 - State Boundary
 - - - District Boundary
 - - - Block Boundary
 - Basin Boundary
 - - - Section Lines
 - Major Streams / River
 - Pond / Reservoir
 - Hills
- Exploratory Wells**
- Kod Well location for Sections
 - CGWB
 - RGWD



CROSS SECTIONS

OTHER NALLAHS BASIN

Section A-A':

This section is the longest of the sections plotted in the area and trends in W-E direction, cutting across the basin and covering a distance of about 45 kms (Plate – XIX). Lithologs acquired from 5 boreholes was utilized in interpretation of sub-surface lithology in this section. The section depicts the disposition of different layers of sand and clay along with shale. On perusal of the cross section, it is apparent that the total area along the cross-section is occupied by sand. A narrow strip of clay is also present. Hardrock is encountered in only easternmost well.

The water level varies from 50 m amsl to 140 m amsl in the section following the surface topography.

Section B-B':

The B-B' section trending from NW-SE and covers an area of 18 kms shown in (Plate – XX). The lithologs of 4 boreholes were taken in to account while preparing the section. The section depicts sand and clay layers of alluvial aquifers. The alternate sequence of sand and clay is forming the major aquifer system in these areas.

It is observed from pre monsoon 2010 data the ground water level varies from 50 m amsl to 70 m amsl from NW – SE of the section drawn in the Other Nallahs river basin.

PLATE - XIX

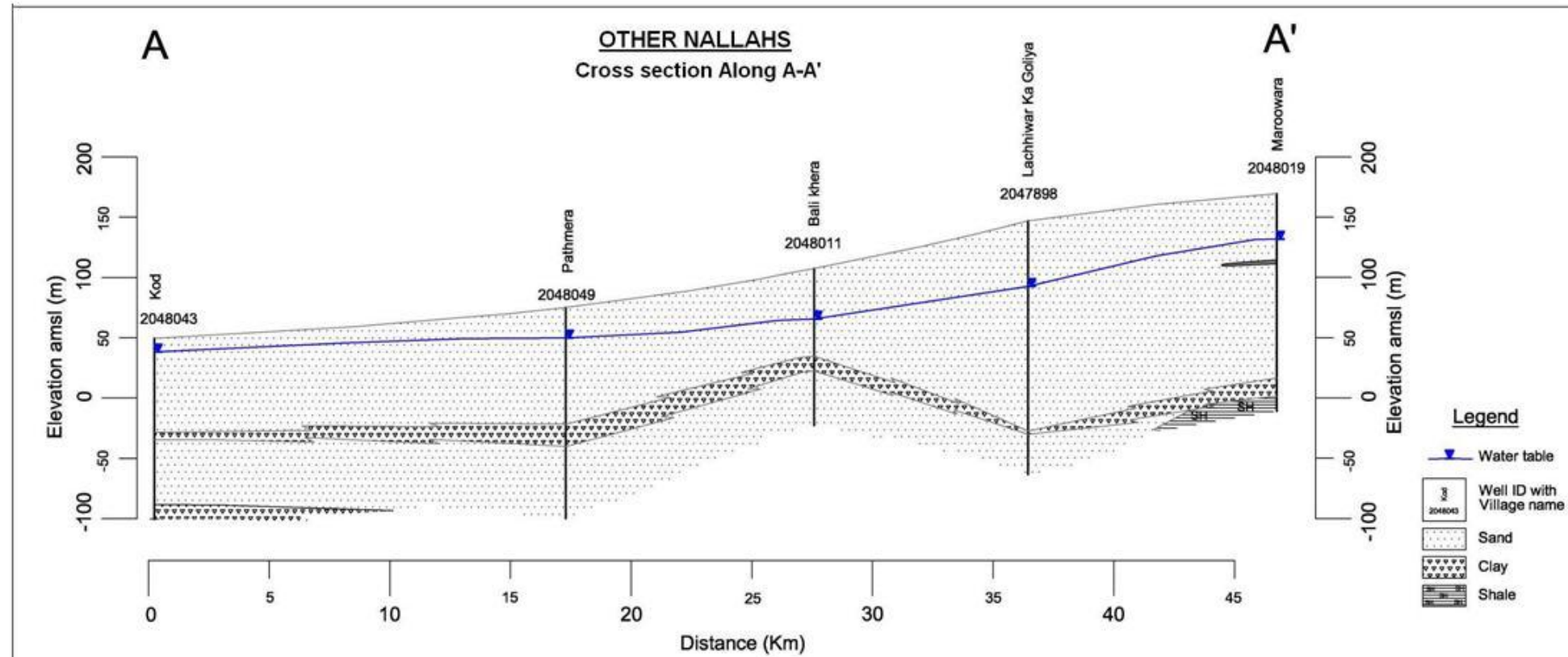
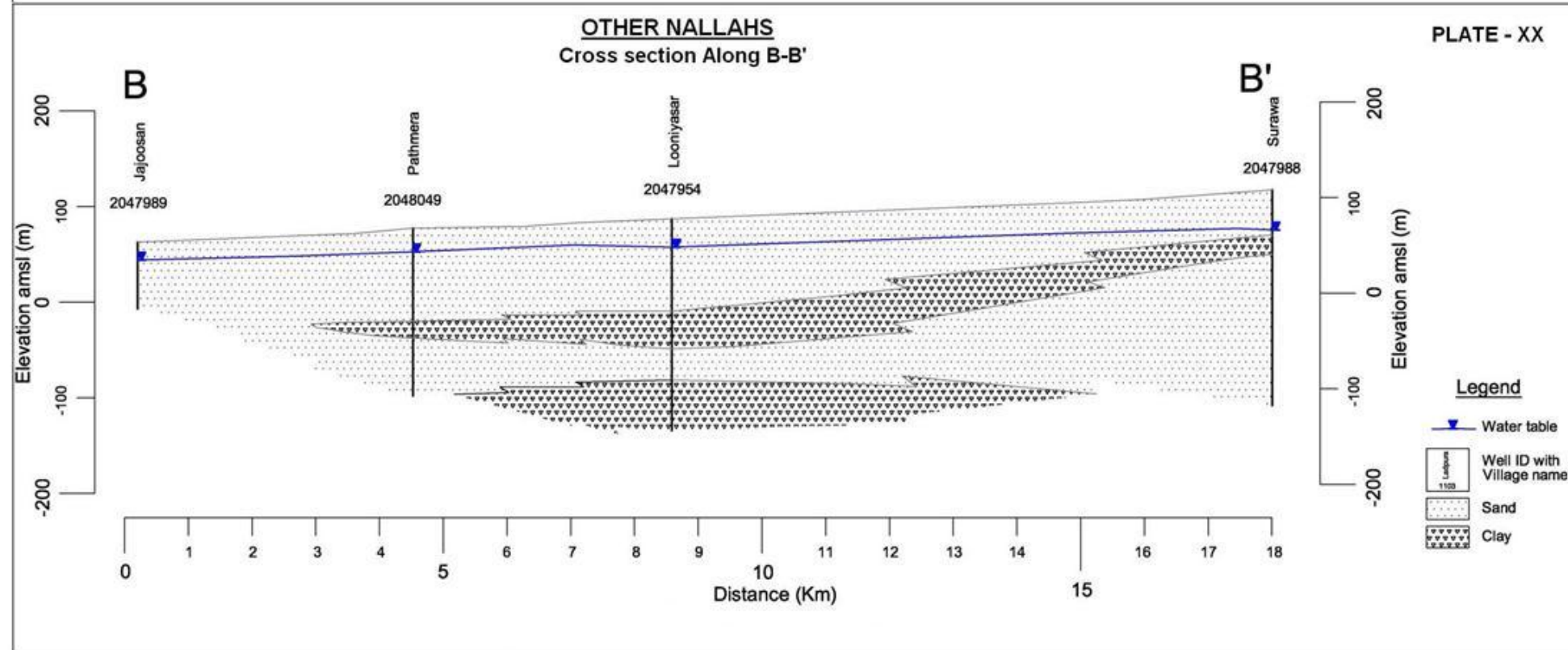


PLATE - XX



CROSS SECTIONS

Section C-C':

The C-C' section has been prepared to visualize variation across the basin trending from NW-SE in the central part of the basin and covers a length of 25 kms (Plate – XXI). The lithologs of 4 boreholes are taken into account while preparing this cross-section. The section depicts the disposition of different layers of sand and clay along with shale. In the rest of the area, sand is predominating while there are some lenses of clay. Hardrock is encountered in easternmost well. The cross-section depicts that alluvium is forming the major aquifer system underlain by shale.

The water level varies from 48 m amsl to 130 m amsl based on the data of pre monsoon 2010.

Section D-D':

The section is trending from NW-SE covers a length of 23 kms (Plate – XXII). To depict the lithology of the cross-section, 4 boreholes along with surrounding well information are chosen. The cross section cuts across two streams namely Bargaon and Sukal. The section depicts the disposition of sand along with granite. Granite comprises the bedrock which is overlain by sand. The thickness of sand varies from 5m to 65m at various places in the section

It is observed from pre monsoon 2010 data the ground water level varies from 215 m amsl to 270 m amsl.

OTHER NALLAHS BASIN

PLATE - XXI

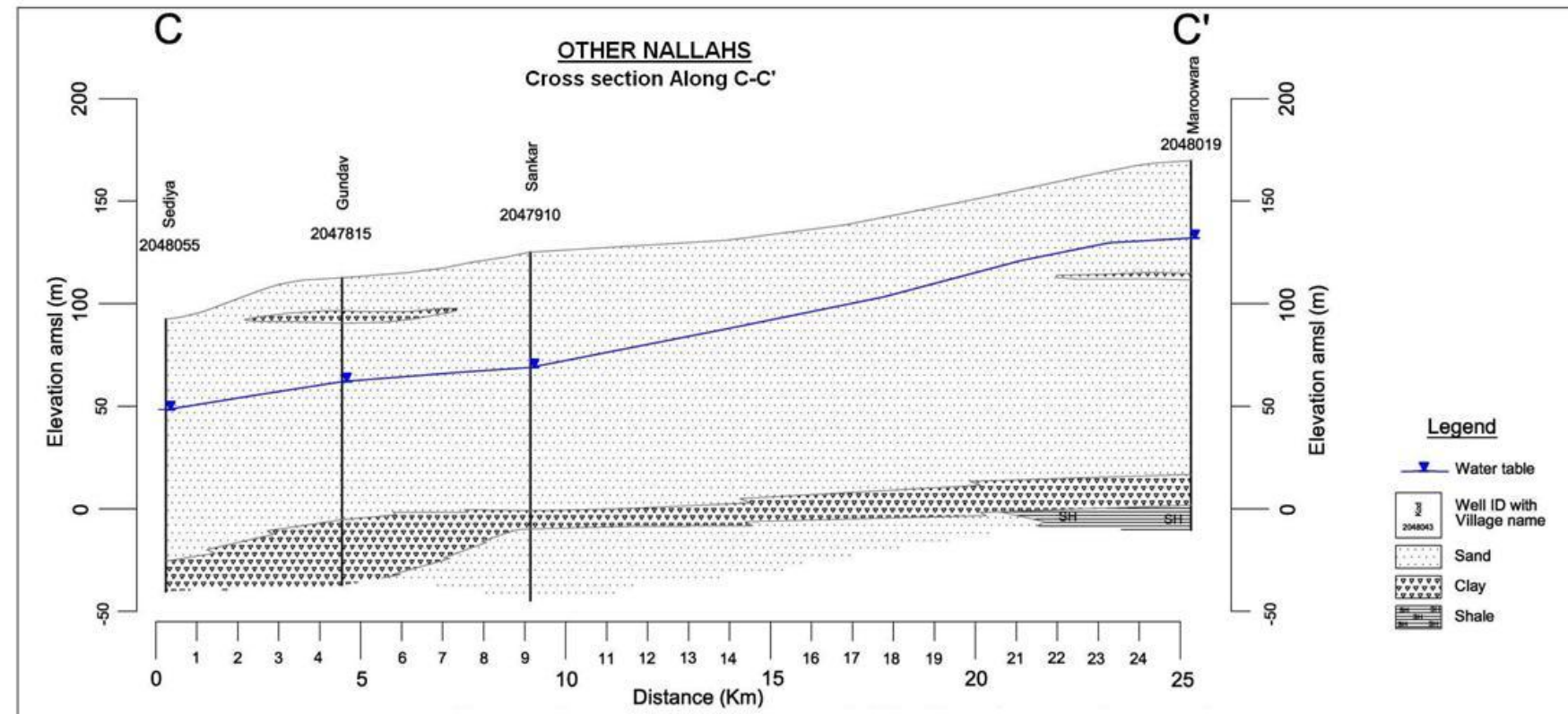
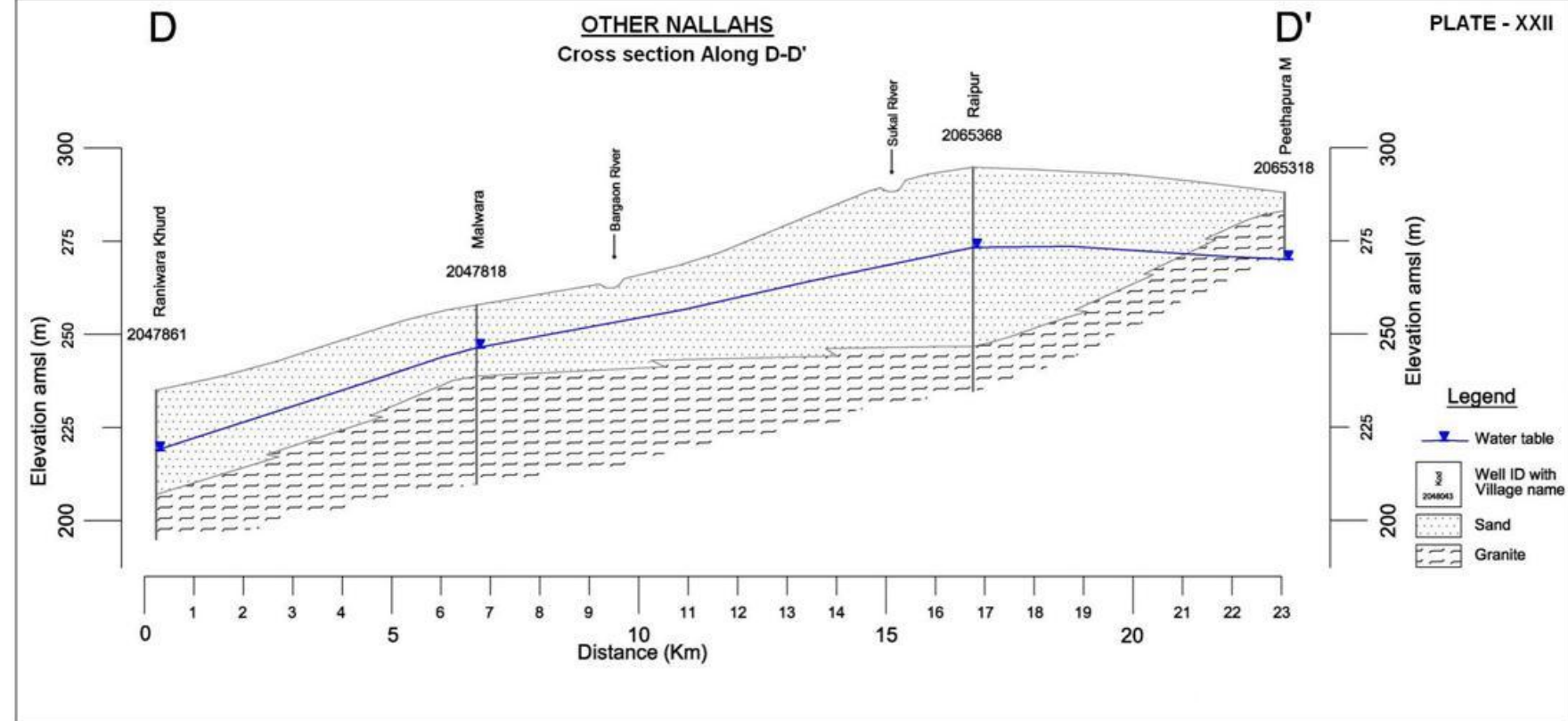


PLATE - XXII



3D MODEL OF AQUIFERS

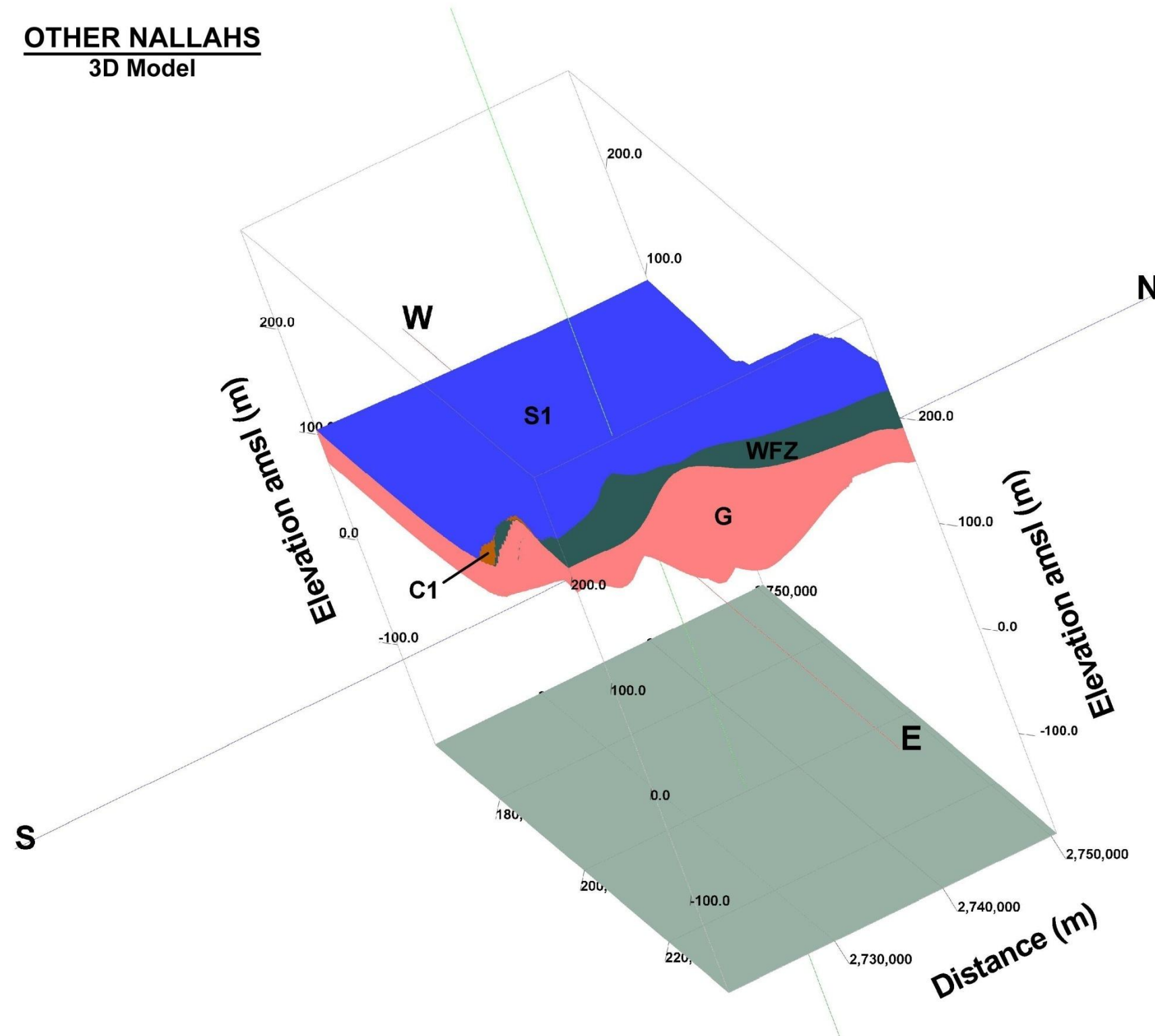
OTHER NALLAHS BASIN

The three dimensional litho-stratigraphic model has been developed for the Other Nallahs basin using interpolation techniques on the data of scattered wells given as an input. Plate – XXII represents a 3D model depicting the various litho-stratigraphic units in the entire river basin. With this model it is apparent that beneath the thin cover of top soil, there are two persistent clay horizons in the region separating two sandy aquifers. The top sandy aquifer is present in the whole area. Beneath the sand layers there is indication of a persistent weathered and fractured bed rock acting as aquifer. From the model it has also been observed that a small patch of weathered and fractured rock acting as an unconfined aquifer in the southwestern part of the area.

The depth of hard rock is shallow in the southeastern part as compared to the northeastern part of the basin. Granite constitutes the basement.

PLATE - XXIII

OTHER NALLAHS
3D Model



Legend

S1	Sand
C1	Clay
S2	Sand
C2	Clay
WFZ	Weathered & Fractured Zone
G	Granite

Glossary of terms

S. No.	Technical Terms	Definition
1	AQUIFER	A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.
2	ARID CLIMATE	Climate characterized by high evaporation and low precipitation.
3	ARTIFICIAL RECHARGE	Addition of water to a groundwater reservoir by man-made activity
4	CLIMATE	The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region.
5	CONFINED AQUIFER	A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.
6	CONTAMINATION	Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.
7	DRAWDOWN	The drawdown is the depth by which water level is lowered.
8	FRESH WATER	Water suitable for drinking purpose.
9	GROUND WATER	Water found below the land surface.
10	GROUND WATER BASIN	A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.
11	GROUNDWATER RECHARGE	The natural infiltration of surface water into the ground.
12	HARD WATER	The water which does not produce sufficient foam with soap.
13	HYDRAULIC CONDUCTIVITY	A constant that serves as a measure of permeability of porous medium.
14	HYDROGEOLOGY	The science related with the ground water.
15	HUMID CLIMATE	The area having high moisture content.
16	ISOHYET	A line of equal amount of rainfall.
17	METEOROLOGY	Science of the atmosphere.
18	PERCOLATION	It is flow through a porous substance.
19	PERMEABILITY	The property or capacity of a soil or rock for transmitting water.
20	pH	Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).
21	PIEZOMETRIC HEAD	Elevation to which water will rise in a piezometers.
22	RECHARGE	It is a natural or artificial process by which water is added from outside to the aquifer.
23	SAFE YIELD	Amount of water which can be extracted from groundwater without producing undesirable effect.
24	SALINITY	Concentration of dissolved salts.
25	SEMI-ARID	An area is considered semiarid having annual rainfall between 10-20 inches.
26	SEMI-CONFINED AQUIFER	Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.
27	SPECIFIC YIELD	Quantity of water which is released by a formation after its complete saturation.
28	TOTAL DISSOLVED SOLIDS	Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.

(Contd...)

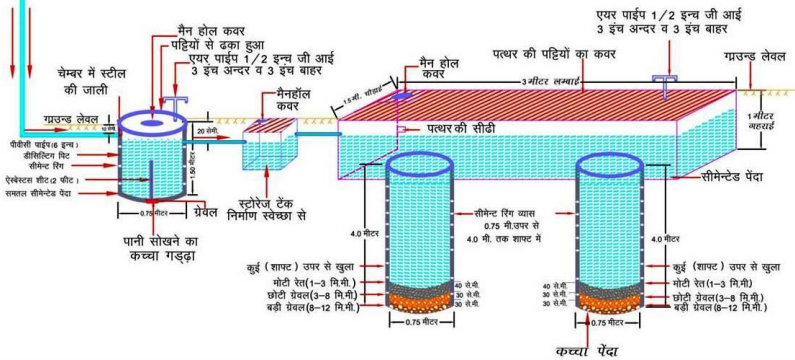
S. No.	Technical Terms	Definition
29	TRANSMISSIBILITY	It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability.
30	UNCONFINED AQUIFER	A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.
31	UNSATURATED ZONE	The zone below the land surface in which pore space contains both water and air.
32	WATER CONSERVATION	Optimal use and proper storage of water.
33	WATER RESOURCES	Availability of surface and ground water.
34	WATER RESOURCES MANAGEMENT	Planned development, distribution and use of water resources.
35	WATER TABLE	Water table is the upper surface of the zone of saturation at atmospheric pressure.
36	ZONE OF SATURATION	The ground in which all pores are completely filled with water.
37	ELECTRICAL CONDUCTIVITY	Flow of free ions in the water at 25C mu/cm.
38	CROSS SECTION	A Vertical Projection showing sub-surface formations encountered in a specific plane.
39	3-D PICTURE	A structure showing all three dimensions i.e. length, width and depth.
40	GWD	Ground Water Department
41	CGWB	Central Ground Water Board
42	CGWA	Central Ground Water Authority
43	SWRPD	State Water Resources Planning Department
44	EU-SPP	European Union State Partnership Programme
45	TOPOGRAPHY	Details of drainage lines and physical features of land surface on a map.
46	GEOLOGY	The science related with the Earth.
47	GEOMORPHOLOGY	The description and interpretation of land forms.
48	PRE MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)
49	POST-MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)
50	PIEZOMETER	A non-pumping small diameter bore hole used for monitoring of static water level.
51	GROUND WATER FLUCTUATION	Change in static water level below ground level.
52	WATER TABLE	The static water level found in unconfined aquifer.
53	DEPTH OF BED ROCK	Hard & compact rock encountered below land Surface.
54	G.W. MONITORING STATION	Dug wells selected on grid basis for monitoring of state water level.
55	EOLIAN DEPOSITS	Wind-blown sand deposits



भवन छत क्षेत्रफल 300 से 500 वर्गमीटर तक
निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

चित्र-4

- PVC पाईप 6" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.50 मी गहरा)
- रीचार्ज टैंक 1.5 मी चौड़ा x 3 मी लम्बा x 1 मी गहरा
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा) (संख्या 2)
- संरचना की अनुमानित लागत रु 24,000 अधिक
- वार्षिक पुनर्भरित जल लगभग 2,00,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 40,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



भूजल में घुले मुख्य तत्वों की अधिकता का मानव शरीर पर दुष्प्रभाव

बोरॉन-स्नायु तन्त्र पर प्रभाव

फ्लोराइड - दंत क्षरण

क्लोराइड-सोडियम के साथ मिलकर उच्च रक्त चाप

सोडियम-हृदय, गुर्दा व रक्त परिसंचरण रोगों से ग्रसित लोगों को हानिकारक

कैल्शियम-जोड़ों में कड़ापन

नाइट्रेट-नवजात शिशुओं में ब्लू बेबी बीमारी (मेथेमोग्लोबिनिमिया)

आर्सेनिक-त्वचा रोग, कैंसर

सल्फेट-अधिकता में मैग्नेशियम के साथ मिलकर दस्तावर

लेड-बच्चों के शारीरिक व मानसिक विकास में बाधा वयस्कों में गुर्दे के रोग

आयरन-आयरन जीवाणु से आमाशय संबंधी रोग

फ्लोराइड-जोड़ों में अकड़न, हड्डियों में मुड़ाव



केन्द्रीय भूमि जल बोर्ड,
पश्चिमी क्षेत्र, जयपुर
जल संसाधन मंत्रालय
भारत सरकार
e-mail: cgwbwr@sancharnet.in



भूजल अमूल्य है इसे प्रदूषित न करें।

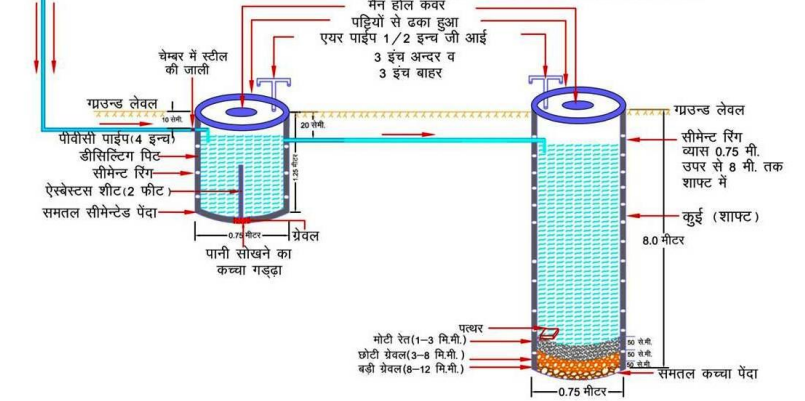


भवन छत क्षेत्रफल 100 से 200 वर्गमीटर तक

चित्र-2

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

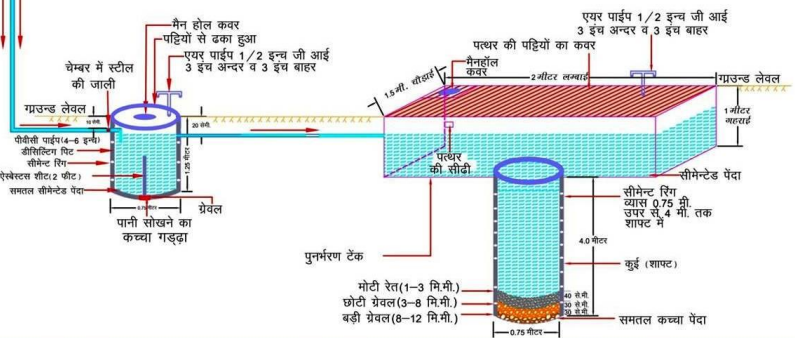
- PVC पाईप 4" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत रु 11,000-12,000
- वार्षिक पुनर्भरित जल लगभग 83,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 16,64,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



भवन छत क्षेत्रफल 200 से 300 वर्गमीटर तक
निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

चित्र-3

- PVC पाईप 4" - 6" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- रीचार्ज टैंक 1.5 मी चौड़ा x 2 मी लम्बा x 1 मी गहरा
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत रु 15,000-16,000
- वार्षिक पुनर्भरित जल लगभग 1,25,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 25,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम

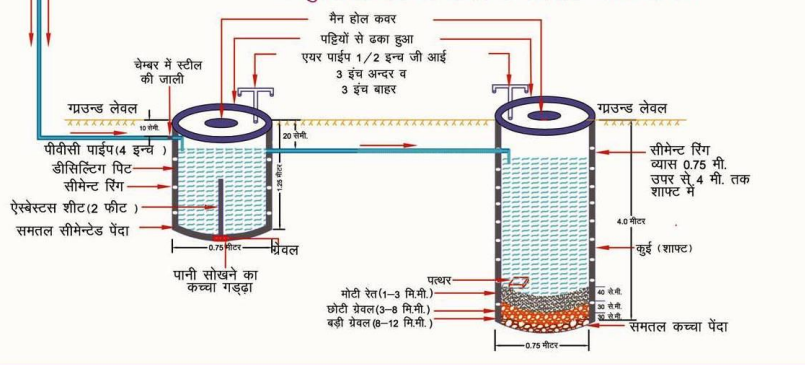


भवन छत क्षेत्रफल 100 वर्गमीटर तक

चित्र-1

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

- PVC पाईप 4" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत रु 7000-8000
- वार्षिक पुनर्भरित जल लगभग 40,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 8,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



Myths and Facts about Ground Water

S No	Myths	Facts
1	What is Ground Water • an underground lake • a net work of underground rivers • a bowl filled with water	Water which occurs below the land in geological formations/rocks is Ground water
2	Ground Water occurs everywhere beneath the Land Surface	Not really, it depends on the nature of rock formation
3	There is a relationship between ground water and surface water	Not all the places. Near streams/rivers there is relation
4	Groundwater is not renewable resource	It is renewable source and every year it is being recharged through rain/applied irrigation etc
5	Ground water is unlimited and deeper you drill more discharge	It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper
6	Ground Water moves rapidly	The movement of ground water is very slow
7	Ground water pumped from wells is thousands of years old	Generally the ground water being tapped through wells is a few years old
8	If water taste good—it is safe to drink	It may have other chemicals e.g. fluoride, nitrates etc which are harmful
9	Water from free flowing tube wells is very pure	This water can also be contaminated so test before use
10	If I recharge my TW/DW/HP it will not benefit me	It will also benefit you and also adjoining wells
11	There is no static ground water resources in Rajasthan	Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge
12	I cannot meet annual cooking and drinking water requirement by rain water harvesting	The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement
13	You can increase ground water recharge	This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures
14	You cannot use abandoned TW/HP/DW for ground water recharge	These should be used as recharge structures as harvested rain water is directly put into GW reservoir
15	Putting waste near HP/TW will not cause any problem	Such actions will pollute wells and water



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